Advanced Combustion via Microgravity Experiments (ACME)

Data Camera Characterization Report

Rev. A
August, 2016





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PREFACE

Under the Space Flight Systems Development and Operations Contract (SpaceDOC), the National Aeronautics and Space Administration (NASA) has developed a modular, multi-user experimentation facility for conducting fluid physics and combustion science experiments in the microgravity environment of the International Space Station (ISS). This facility, called the Fluids and Combustion Facility (FCF), consists of two test platforms: the Fluids Integrated Rack (FIR) and the Combustion Integrated Rack (CIR). Also included in SpaceDOC are the required support efforts for Mission Integration and Operations, consisting of the Telescience Support Center (TSC) and Mission Integration and Planning (MIP).

The Advanced Combustion via Microgravity Experiments (ACME) is mini-facility designed to support Principal Investigators (PI) interested in studying various aspects of advanced combustion in a microgravity environment. ACME will be integrated into the Combustion Integrated Rack (CIR) onboard the International Space Station in the US Lab Module.

ACME will perform microgravity experiments to study fire suppression/flame extinguishment via 5 experiments: s-flame, Flame design, CLD Flame, E-Field Flames, and Burning Rate Emulator Flames as described in the ACME Baseline Description Document, ACME-DOC-004. ACME will allow investigation using pure as well as mixtures of gaseous fuels at various chamber atmosphere pressures, compositions, and oxygen concentrations. The microgravity environment allows for observation and measurement of flame formation and extinguishment processes that cannot be made in sustained gravity. The modular design of ACME allows for flexibility in configuration for specific experiments and easy maintainability of the hardware.

The core elements of gaseous combustion are included in the ACME design as a basis for current and future gaseous combustion studies. These elements are establishment of gas flow through various exchangeable mass flow controllers, ignition of gas at the burner and the use of appropriate diagnostic measurement devices. The five identified experiments, and any future experiments, use these core elements to build a unique set of requirements for the study of various phenomena distinctive to gaseous combustion. The goal of ACME is to improve the fundamental understanding of combustion processes in order to enable increased efficiency and reduce pollutant emission in practical combustion.

SIGNATURES / APPROVALS

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REVISIONS

Revision Letter	Publication Date	Change Description	Approval
-	08/2016	Baseline Release	-
A	11/2016	Per Science request, Blackbody and Flame Thermocouple tests were repeated with different configurations.	BB 11/16/2016 EM 11/16/2016

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1.0 INTRODUCTION

1.1 Purpose

The purpose of this report is to document the results for the characterization of the ACME Data Camera Package (S1101MFA2000), Flight unit, S/N 1002.

1.2 Scope

The scope of this report covers all characterizations, calibrations and tests that were performed in May, 2016, along with additional characterizations completed in October 2016, on the ACME flight model. Results and subsequent analysis will be recorded in this final characterization report. This document does not cover all verification test results, but may be used as a reference for verifications.

1.3 Classification

The classification for the ACME Data Camera Package is Flight Model, including all the hardware listed in Table II.

2.0 DOCUMENTS

This section lists specifications, models, standards, guidelines, handbooks, and other special publications. These documents have been grouped into two categories: applicable documents and reference documents.

2.1 Order of Precedence for Documents

In the event of a conflict between this document and other documents referenced herein, the information of this document shall apply. In the event of a conflict between this document and the contract, the contractual requirements shall take precedence over this document. All documents used, applicable or referenced are to be the issues defined in the Configuration Management (CM) contract baseline. All document changes, issued after baseline establishment, shall be reviewed for impact on scope of work. If a change to an applicable document is determined to be effective and contractually approved for implementation, the revision status will be updated in the CM contract baseline. The contract revision status of all applicable documents is available by accessing the CM database. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

2.2 Applicable Documents

The following documents are applicable to the ACME project to the extent specified herein.

S1101MFA2000	ACME Data Camera Assembly
ACME-REQ-001 Rev. J	ACME Engineering Requirements Document
ACME-TST-019 Rev. C	Data Camera Control Enclosure Test Procedure
ACME-TST-061 Rev. -	Data Camera Characterization Test Plan & Procedure
P13001	SpaceDOC Procedure for the Control of Nonconforming Product
P05002	Configuration Management Procedure
P30020	SpaceDOC Safety and Mission Assurance Plan

2.3 Reference Documents

The documents/drawings in this section are provided only as reference material for background information and are not imposed as requirements.

ANSI-ESD S20.20.2007	ESD Association for the Development of an Electrostatic Discharge Program for Protection of Electronic Parts, Assemblies and Equipment (excluding Electrically Initiated Explosive Devices)
ACME-DOC-002 Rev. D	ACME Compliance Matrix
ACME-DOC-007 Rev. A	ACME Concept of Operations
ACME-SRD-001 Rev. -	Integrated Science Requirements Document

3.0 FACILITIES, EQUIPMENT, AND INSTRUMENTATION

3.1 Test Facilities

The ACME flight Data Camera Package Characterization took place in May, 2016 and October 2016 in the ZIN Technologies lab facility located at 6749 Engle Road.

3.2 Equipment and Instrumentation

The following table lists the name and type of the equipment and instrumentation required for the ACME Data Camera Package Characterization. The calibration status of all instrumentation was verified as appropriate before the characterization began.

TABLE I. Equipment and Instrumentation Information

Equipment	Manufacturer	Model Number	Cal Due Date
CIR UML Simulator	ZIN Technologies	67212ETAH20820 S/N: 001-B, NASA Tag: 25556	N/A
CIR Optics Bench Simulator	ZIN Technologies	67212MGAF16201 NASA Tag: 06895	N/A
CIR Sapphire Window Assembly	ZIN Technologies	67212MEDH33000 S/N: 2006	N/A
ACME GSE Test Rack	ZIN Technologies	S1101MGA4500 S/N: 56801	N/A
ACME Avionics Package (AVP)	ZIN Technologies	S1101MFA3000 S/N: 1001	N/A
Dual DC Power Supply, +3.3 VDC / 24 VDC	Tektronix	Model: PS-280 S/N: TW52908, NASA Tag: 12129	N/A (via voltmeter)
DC Regulated Power Supply, 28 VDC	Kenwood	Model: PD36-20AD S/N: 5010016, ZIN Tag: Z000162	N/A (via voltmeter)
Digital Voltmeter/Multi-meter	Fluke	Model: 179 S/N: 81730258, ZIN Tag: Z000371	11 May, 2016
GSE Test Harness	ZIN Technologies	S1101EGA4552 S/N: 1001	N/A
Data Camera Test Harness & AVP to DataCam Harness	ZIN Technologies	S1101EGA4533, S110EFA2002 S/N: 1001	N/A
Laptop w/ Ethernet & Serial Port AVI Vimba (≥ Ver.2.1) Pro-Motion SW (≥ Ver.3.83)	Dell	Model: Latitude E5510 S/N: CR8W4N1 ZIN Tag: Z004622	N/A
Integrating Sphere (20”) w/ Controller & Power Supply	Labsphere	Model: SC-5500 S/N: 06561, NASA Tag: 13309	N/A
LED Light Source w/ LEDs (White Light, 455nm)	Lightspeed Technologies	Model: HPLS-36APL750 S/N: 0609520, NASA Tag: 43895	N/A
Blackbody Infrared Source	Infrared Systems Development	Model: IR-564 S/N: 3062K, NASA Tag: 09083	7/2016
Blackbody Temp. Controller	Infrared Systems Development	Model: IR-301 S/N: 3062K, NASA Tag: 09083	7/2016
Torch Assembly: Mag-Torch Burner 4” Extension Hose Searzall Torch Diffuser 16.4 oz Propane Canister		N/A	N/A

4.0 DESCRIPTION OF TEST ARTICLE & CONFIGURATION

4.1 Test Article

4.1.1 ACME Data Camera Assembly

The ACME Data Camera Assembly (S1101MFA2000) consists of three sub-assembly components that make up the entire assembly: the Camera Control Assembly (S1101MFA2100), the Filter Barrel Assembly (S1101MFA2300) and the Camera Enclosure (S1101MFA2200).

The Data Camera Assembly flight unit was assembled in-house by ZIN Technologies at 6749 Engle Road.

TABLE II. Article

Description	Manufacturer	Drawing Number	S/N
ACME Data Camera Assembly	N/A	S1101MFA2000	1002

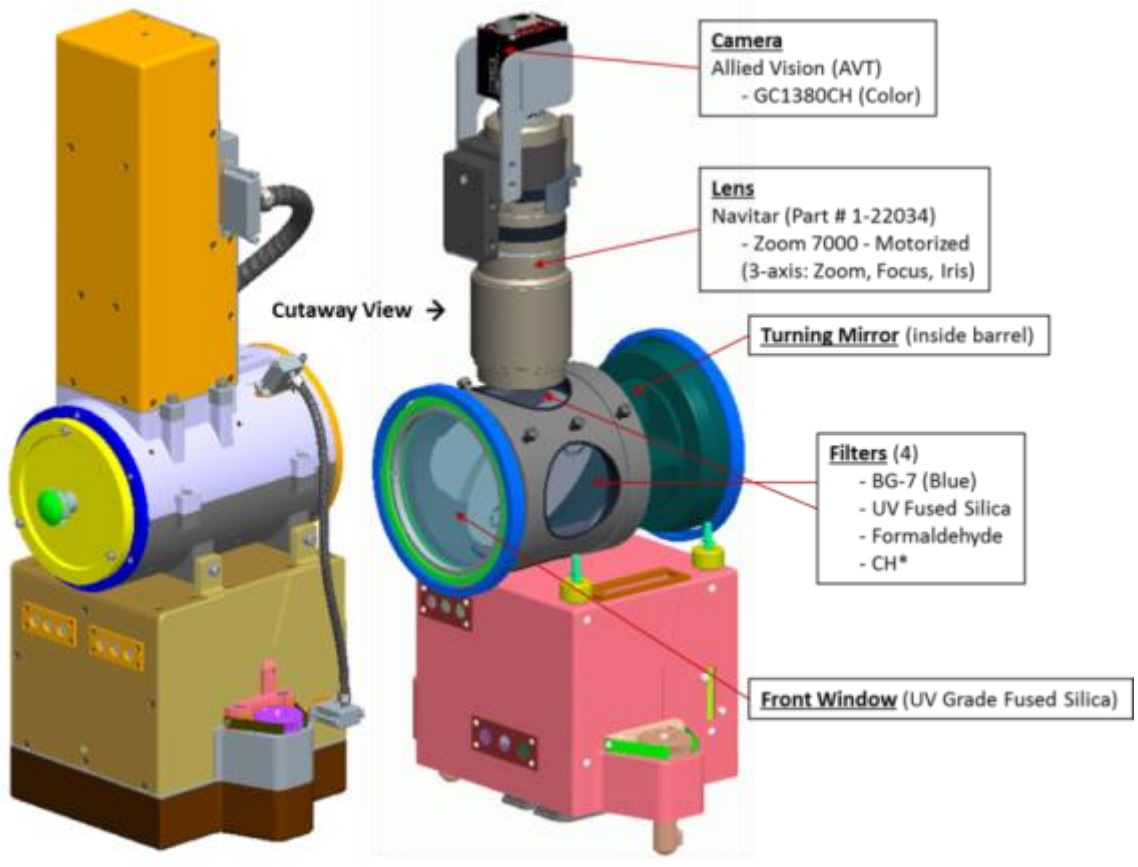


FIGURE 1. ACME Data Camera Assembly

4.1.1.1 Camera Control Module (CCM)

The Camera Control Module (CCM) is shown in Figure 2. It serves as the mounting interface to a CIR Universal Mounting Location (UML). The CCM supplies Power Distribution capability, using 28 VDC input from CIR and converts it to 12 VDC required by the camera and other components. The CCM also routes forced air cooling provided by the CIR up through the internal electronics.

Also housed in the CCM are boards necessary for Stepper Driver and Motion Control functions that are used by the rotating filter barrel and the lens for zoom, focus and iris functions. Finally RS-232 and RS-422 communications capabilities are supported by boards housed here as well.



FIGURE 2. Camera Control Module (CCM)

4.1.1.2 Filter Barrel Assembly

The front end of the Filter Barrel Assembly (Figure 3) is a UV Fused Silica front Window for protection of the internal components. The back end is a 45° turning Mirror with a UV enhanced Al coating, in order to fold the optical path.

Within the Filter Barrel Assembly is a stepper motor driven filter barrel with rotational capability. The barrel itself has slots available for 4 individual filters.

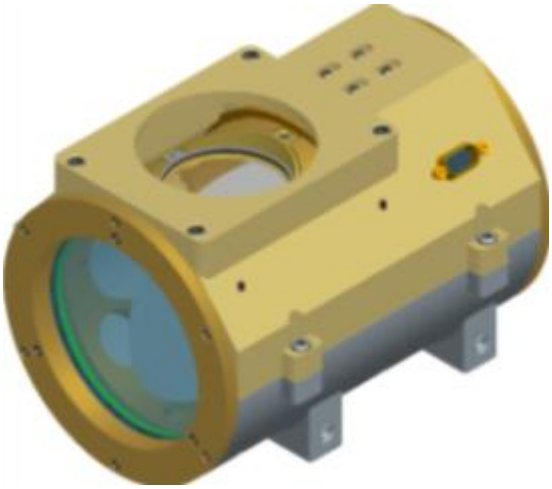


FIGURE 3. Filter Barrel Assembly

ACME has selected the following for the 4 slots: 1 – CH* emissions (431 nm ± 10 nm), 2 – UV Fused Silica, 3 – Formaldehyde emissions (390 nm – 490 nm) and 4 – Schott BG-7. Transmission curves for the various filters are presented in Section 5.9– Spectral Information.

4.1.1.3 Camera Enclosure

The Camera Enclosure includes the base camera (Allied Vision model GC-1380HC), and Motorized Zoom Lens (Navitar 7000).

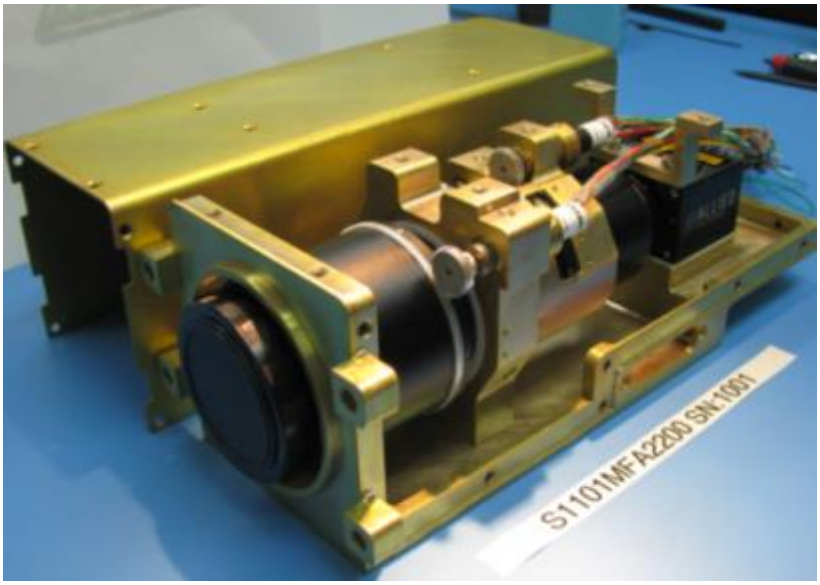


FIGURE 4. Camera Enclosure

4.1.1.4 Test Configuration

For this Characterization, the ACME Data Camera Assembly was mounted on a CIR UML Simulator tub and positioned on an optical bench. The outer window of the assembly was aligned with the opening of the Integrating Sphere. A CIR Window Assembly (Sapphire) was mounted directly in front of the Data Camera Assembly in order to replicate on-orbit conditions.

Various targets are used as indicated throughout the characterization and are mounted via additional optical mounting hardware that allows for alignment in both the X and Y directions and also sets the distance from the outer window surface of the Data Camera Assembly to the target (Z direction), equal to the appropriate CIR flight distance of 253.2 mm.

All testing was done under blackout curtains that were arranged such that only the light from the Integrating Sphere was allowed to enter during characterization. See the picture below for the full setup (Note: side curtains are raised in order to obtain the picture.)

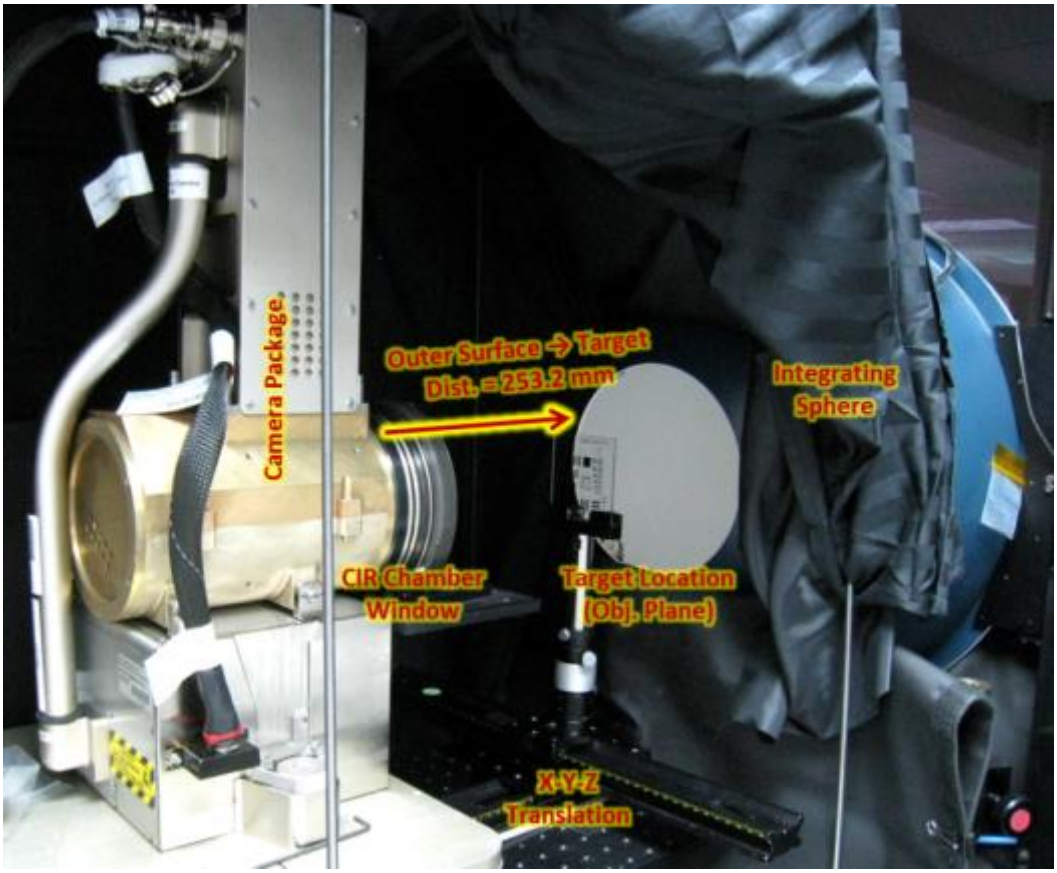


FIGURE 5. Data Camera Assembly Alignment Setup

The CIR UML Simulator tub provides power and fan cooling to the package. The UML Simulator requires 2 power supplies: Fan requires 24 VDC, while the Package requires 28 VDC. These are connected via the UML Simulator Power Harness.



FIGURE 6. GSE for Characterization Setup

Camera data output and commanding is connected to a laptop computer via the Camera Test Harness with Breakout Panel, which provides both a serial connection for motor control, as well as an Ethernet connection for imaging capability. Refer to the connection schematic below.

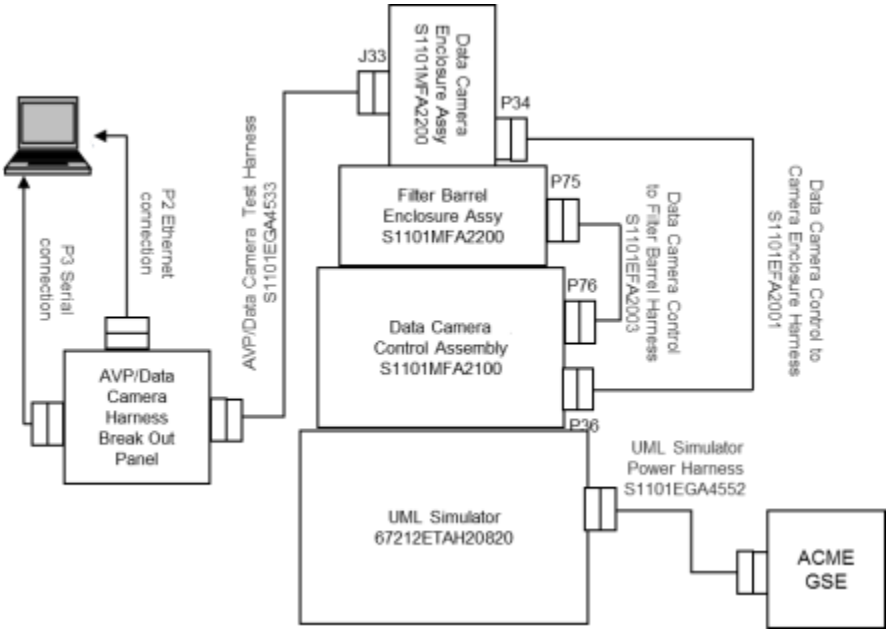


FIGURE 7. Cabling Schematic for Characterization Setup

5.0 CHARACTERIZATION RESULTS

5.1 Dark Frame Images and Gain

5.1.1 Setup

The Data Camera Package was configured (as shown in FIGURE 5) with the Iris closed completely (motor maximum counts = 75,900) in order to block incoming light. In addition the front lens cover was installed in order to further reduce the chance of light reaching the CCD.

Specific camera settings for this section are as follows (grayed out indicates not applicable):

AVT Camera Functions			Motor Control Functions		
Width/Height (# pixels)	Width	1360	Filter(s) Used	CH*	10,000
	Height	1024		Fused Silica	120,000
Full Frame or Binning	Horiz	1x		Formaldehyde	235,000
	Vert	1x		BG-7	350,000
White Balance	R / B	0.9 / 3.0	Iris	Closed	75,900
Gain	db	0	FOV (mm)	Zoom	Focus
Exposure Time	Full Frame @ 30 fps	32,000	54.9 x 41.3	20,000	14,750
Pixel Format (Bit Depth)	12 Bit - Raw	BayerRG12	141.6 x 106.6	55,000	6,900

All images are in the Raw format of the camera (**.bin** file extension), unless otherwise specified.

5.1.2 Data / Analysis

10 images were taken at with the Gain = 0. These images were analyzed, using ImageJ, both individually and also as a 10-image "stack" resulting in average output. This analysis was over the full 12-bit image (GL from 0 – 4095), providing Min & Max Gray Levels, along with the Mean Gray Level (GL) and Standard Deviation. The resulting data is shown below.

12-Bit Raw @ Gain = 0	Min	Max	Mean	StdDev
Dark Frame Gain 0_1	2	249	53.19	33.5
Dark Frame Gain 0_2	0	273	53.68	34.7
Dark Frame Gain 0_3	0	282	53.57	34.6
Dark Frame Gain 0_4	1	306	53.10	33.4
Dark Frame Gain 0_5	0	246	53.63	34.6
Dark Frame Gain 0_6	2	273	53.15	33.5
Dark Frame Gain 0_7	0	297	53.58	34.6
Dark Frame Gain 0_8	0	270	53.18	33.5
Dark Frame Gain 0_9	1	291	53.58	34.6
Dark Frame Gain 0_10	2	264	53.19	33.5
Average (12-Bit)	0.8	275.1	53.39	34.1
Average (Normalized)	0.0	0.067	0.013	0.008
Stack Average	24	275	52.94	32.8

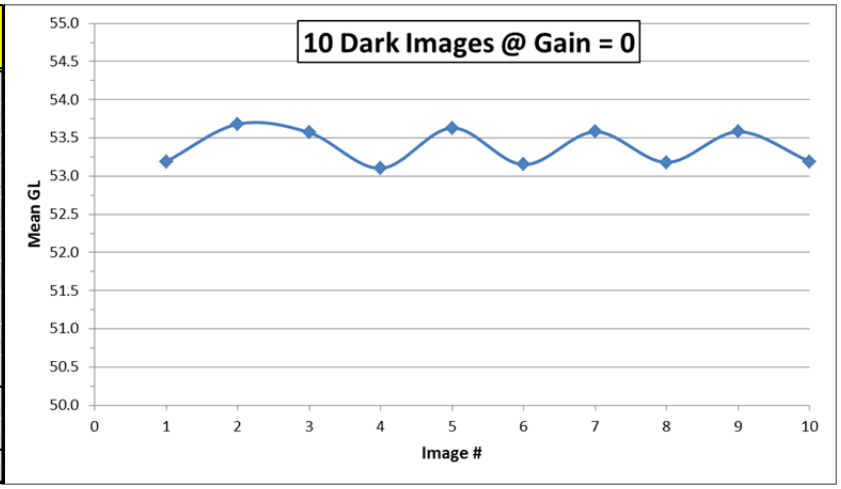


FIGURE 8. Dark Frames: Average of Mean GL at Gain = 0

In order to further characterize the Gain, 10 images were taken at Gain settings of 5, 10, 15, 20, 25, 30 and 33 (max value). These images were analyzed, using ImageJ, both individually and also as a 10-image "stack" resulting in average output. This analysis was over the full 12-bit image (GL from 0 – 4095), providing Min & Max Gray Levels, along with the Mean Gray Level (GL) and Standard Deviation. The resulting data and associated curve is shown below.

TABLE III. Dark Frames Data

12-Bit Raw Gain = 0	Min	Max	Mean	StdDev
Dark Frame Gain 0_1	2	249	53.19	33.5
Dark Frame Gain 0_2	0	273	53.68	34.7
Dark Frame Gain 0_3	0	282	53.57	34.6
Dark Frame Gain 0_4	1	306	53.10	33.4
Dark Frame Gain 0_5	0	246	53.63	34.6
Dark Frame Gain 0_6	2	273	53.15	33.5
Dark Frame Gain 0_7	0	297	53.58	34.6
Dark Frame Gain 0_8	0	270	53.18	33.5
Dark Frame Gain 0_9	1	291	53.58	34.6
Dark Frame Gain 0_10	2	264	53.19	33.5
Average (12-Bit)	0.8	275.1	53.39	34.1
Average (Normalized)	0.0	0.067	0.013	0.008
Stack Average	24	275	52.94	32.8

12-Bit Raw Gain = 5	Min	Max	Mean	StdDev
Dark Frame Gain 5_1	0	423	52.98	37.8
Dark Frame Gain 5_2	0	414	52.12	35.8
Dark Frame Gain 5_3	0	384	52.28	36.0
Dark Frame Gain 5_4	0	489	52.93	37.8
Dark Frame Gain 5_5	0	372	52.22	35.9
Dark Frame Gain 5_6	0	387	52.28	35.9
Dark Frame Gain 5_7	0	369	53.01	37.8
Dark Frame Gain 5_8	0	384	52.18	35.9
Dark Frame Gain 5_9	0	408	53.07	37.9
Dark Frame Gain 5_10	0	351	52.34	36.0
Average (12-Bit)	0	398.1	52.54	36.7
Average (Normalized)	0.0	0.097	0.013	0.009
Stack Average	19	397	52.09	32.8

12-Bit Raw Gain = 10	Min	Max	Mean	StdDev
Dark Frame Gain 10_1	0	714	50.575	42.9
Dark Frame Gain 10_2	0	702	52.015	45.8
Dark Frame Gain 10_3	0	729	50.553	42.9
Dark Frame Gain 10_4	0	651	51.98	45.7
Dark Frame Gain 10_5	0	597	51.841	45.7
Dark Frame Gain 10_6	0	687	50.67	43.0
Dark Frame Gain 10_7	0	669	51.834	45.7
Dark Frame Gain 10_8	0	561	50.717	42.9
Dark Frame Gain 10_9	0	624	51.817	45.6
Dark Frame Gain 10_10	0	531	50.312	42.8
Average (12-Bit)	0	646.5	51.23	44.3
Average (Normalized)	0.0	0.158	0.0125	0.011
Stack Average	8	646	50.78	33.6

12-Bit Raw Gain = 15	Min	Max	Mean	StdDev
Dark Frame Gain 15_1	0	1488	49.89	57.8
Dark Frame Gain 15_2	0	1332	52.17	52.0
Dark Frame Gain 15_3	0	1188	50.18	58.1
Dark Frame Gain 15_4	0	1290	51.97	61.9
Dark Frame Gain 15_5	0	1344	49.89	57.8
Dark Frame Gain 15_6	0	1419	49.76	57.7
Dark Frame Gain 15_7	0	1356	51.61	61.7
Dark Frame Gain 15_8	0	1461	52.10	62.1
Dark Frame Gain 15_9	0	1344	50.01	58.0
Dark Frame Gain 15_10	0	1380	51.84	62.0
Average (12-Bit)	0	1360	50.94	58.9
Average (Normalized)	0.0	0.332	0.012	0.014
Stack Average	0	1360	50.49	32.8

12-Bit Raw Gain = 20	Min	Max	Mean	StdDev
Dark Frame Gain 20_1	0	2544	56.613	83.6
Dark Frame Gain 20_2	0	2247	60.132	89.9
Dark Frame Gain 20_3	0	2508	57.239	84.3
Dark Frame Gain 20_4	0	2286	59.957	89.7
Dark Frame Gain 20_5	0	2301	56.298	83.0
Dark Frame Gain 20_6	0	2766	59.88	89.8
Dark Frame Gain 20_7	0	1995	57.033	83.9
Dark Frame Gain 20_8	0	2475	56.867	83.9
Dark Frame Gain 20_9	0	2754	60.674	90.1
Dark Frame Gain 20_10	0	2388	56.822	83.3
Average (12-Bit)	0	2426	58.15	86.1
Average (Normalized)	0.0	0.592	0.0142	0.021
Stack Average	0	2426	57.70	45.3

12-Bit Raw Gain = 25	Min	Max	Mean	StdDev
Dark Frame Gain 25_1	0	4095	73.71	126.7
Dark Frame Gain 25_2	0	3711	79.95	138.1
Dark Frame Gain 25_3	0	4095	80.19	138.8
Dark Frame Gain 25_4	0	4095	74.52	127.7
Dark Frame Gain 25_5	0	4095	79.46	137.9
Dark Frame Gain 25_6	0	4059	75.13	128.0
Dark Frame Gain 25_7	0	3831	79.7	138.3
Dark Frame Gain 25_8	0	4095	75.11	127.9
Dark Frame Gain 25_9	0	4023	80.27	138.3
Dark Frame Gain 25_10	0	3996	75.14	128.1
Average (12-Bit)	0	4010	77.32	133.0
Average (Normalized)	0.0	0.979	0.019	0.032
Stack Average	0	4009	76.87	65.2

12-Bit Raw Gain = 30	Min	Max	Mean	StdDev
Dark Frame Gain 30_1	0	4095	120.53	228.1
Dark Frame Gain 30_2	0	4095	111.01	207.4
Dark Frame Gain 30_3	0	4095	110.68	208.2
Dark Frame Gain 30_4	0	4095	119.12	226.0
Dark Frame Gain 30_5	0	4095	111.20	207.8
Dark Frame Gain 30_6	0	4095	120.77	227.3
Dark Frame Gain 30_7	0	4095	110.70	207.1
Dark Frame Gain 30_8	0	4095	120.05	226.4
Dark Frame Gain 30_9	0	4095	110.96	206.9
Dark Frame Gain 30_10	0	4095	120.77	226.7
Average (12-Bit)	0	4095	115.58	217.2
Average (Normalized)	0.0	1	0.0282	0.053
Stack Average	0	4095	115.13	102.6

12-Bit Raw Gain = 33	Min	Max	Mean	StdDev
Dark Frame Gain 33_1	0	4095	147.35	283.7
Dark Frame Gain 33_2	0	4095	158.87	301.6
Dark Frame Gain 33_3	0	4095	146.01	282.3
Dark Frame Gain 33_4	0	4095	160.42	310.4
Dark Frame Gain 33_5	0	4095	147.68	284.4
Dark Frame Gain 33_6	0	4095	158.36	307.1
Dark Frame Gain 33_7	0	4095	143.80	279.3
Dark Frame Gain 33_8	0	4095	158.26	307.2
Dark Frame Gain 33_9	0	4095	146.11	282.1
Dark Frame Gain 33_10	0	4095	159.99	309.7
Average (12-Bit)	0	4095	152.68	294.8
Average (Normalized)	0.0	1	0.037	0.072
Stack Average	0	4095	152.24	135.9

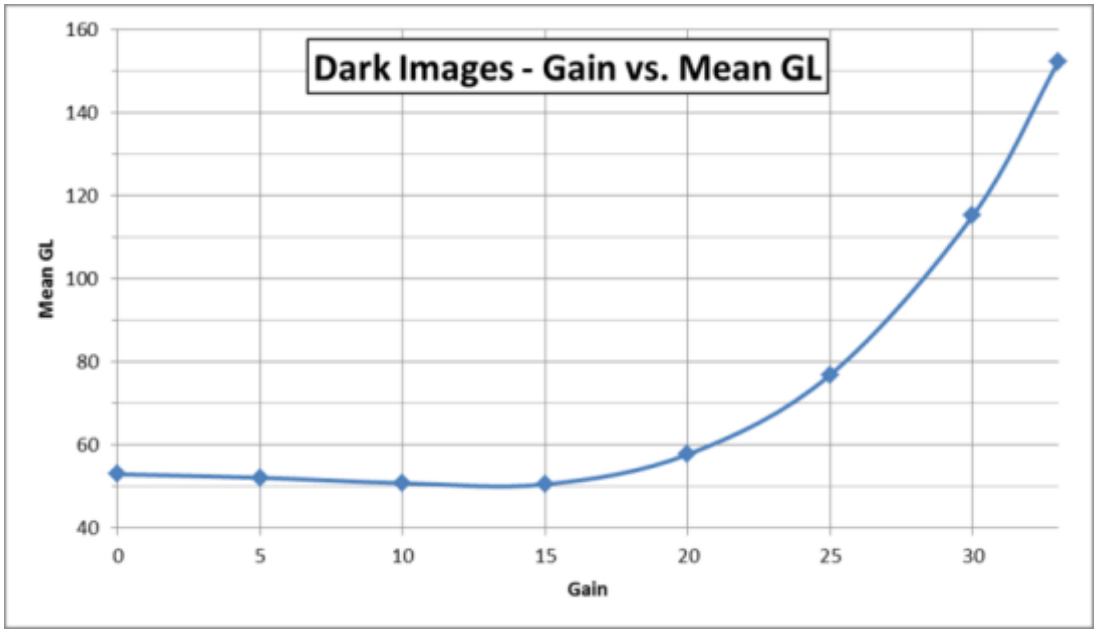


FIGURE 9. Dark Frames: Gain vs. Mean Gray Levels (GL)

5.2 Linearity

5.2.1 Setup

The Data Camera Package was aligned in front of the 20” Integrating Sphere (as shown in FIGURE 5) with a various Neutral Density filters positioned using z-translation, at the specified Object Distance of 253.2 mm from the outer surface of the Data Camera Assembly. The ND filters were centered in the image using x-y adjustments. See an example image of ND Filters below.

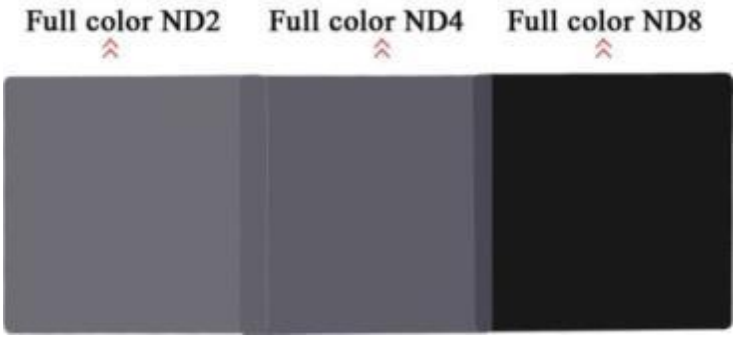


FIGURE 10. Neutral Density Filters

The illumination source used was the QTH Lamp (3000K), with a motorized shutter to vary the output intensity. The illumination was set to maximize the Gray Level without saturating the image for the Fused Silica filter location. This location has the highest transmission and thus would be the first to saturate. Subsequent filters (BG-7 & CH*) were used with the same the illumination setting for a direct comparison. Note that Formaldehyde filter was not characterized due to expected similarity to BG-7 results.

Specific camera settings for this section are as follows (grayed out indicates not applicable):

AVT Camera Functions			Motor Control Functions		
Width/Height (# pixels)	Width	1360	Filter(s) Used	CH*	10,000
	Height	1024		Fused Silica	120,000
Full Frame or Binning	Horiz	1x		Formaldehyde	235,000
	Vert	1x		BG-7	350,000
White Balance	R / B	0.9 / 3.0	Iris	Open	0
Gain	db	0	FOV (mm)	Zoom	Focus
Exposure Time	Full Frame @ 30 fps	Vary	35.3 x 23.8	5,000	15,400
Pixel Format (Bit Depth)	12 Bit - Raw	BayerRG12	Smaller FOV for more Uniform Background		

All images are in the Raw format of the camera (**.bin** file extension), unless otherwise specified.

5.2.2 Data / Analysis

Using the setup described above, once the Illumination source was set, images of the 7 different Neutral Density (ND) filters were captured, along with a "control" image without an ND filter. The ND filters used were: 0.05, 0.2, 0.4, 0.6, 0.8, 2.0 & 3.0. See images below, a subset from the Fused Silica portion of the characterization.

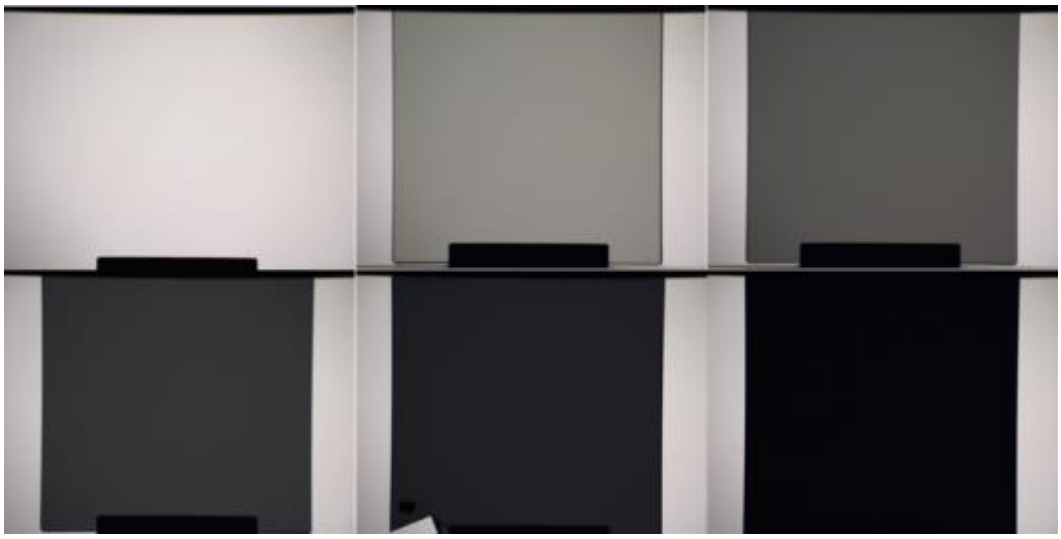


FIGURE 11. Linearity Images (No filter, ND 0.2, ND 0.4, ND 0.6, ND 0.8, ND 2.0)

The set of ND images were recorded for the Fused Silica filter (high end), BG-7 filter (mid-range) and CH* filter (low end). These images were analyzed, using ImageJ, over the Region of Interest (ROI) only covering the ND filter area. Mean Gray Level (GL) was recorded for each image and normalized based on the "No Filter" condition. The resulting data is shown below.

TABLE IV. Linearity Data

Optical	Normalized	F.S.		BG-7		CH*	
Density	Transmission	Mean GL	Relative GL	Mean GL	Relative GL	Mean GL	Relative GL
0	1.0	3404.0	1.000	1252.0	1.000	481.0	1.000
0.05	0.891	3163.0	0.929	1156.0	0.923	449.0	0.933
0.2	0.631	2216.0	0.651	802.0	0.641	289.0	0.601
0.4	0.398	1348.0	0.396	481.0	0.384	192.0	0.399
0.6	0.251	819.0	0.241	305.0	0.244	128.0	0.266
0.8	0.158	513.0	0.151	192.0	0.153	96.0	0.200
2	0.010	64.0	0.019	32.0	0.026	32.0	0.067
3	0.001	48.0	0.014	30.0	0.024	26.0	0.054

The curves resulting from the above data are indeed linear, as expected, especially in both the high end (Fused Silica filter) and the mid-range (BG-7 filter) essentially following the “ideal” linear line. The low end (CH* filter) is also linear but with a slightly different slope than “ideal” linearity. This is typical for the lower end response as the noise floor is approached. See the representative curves below.

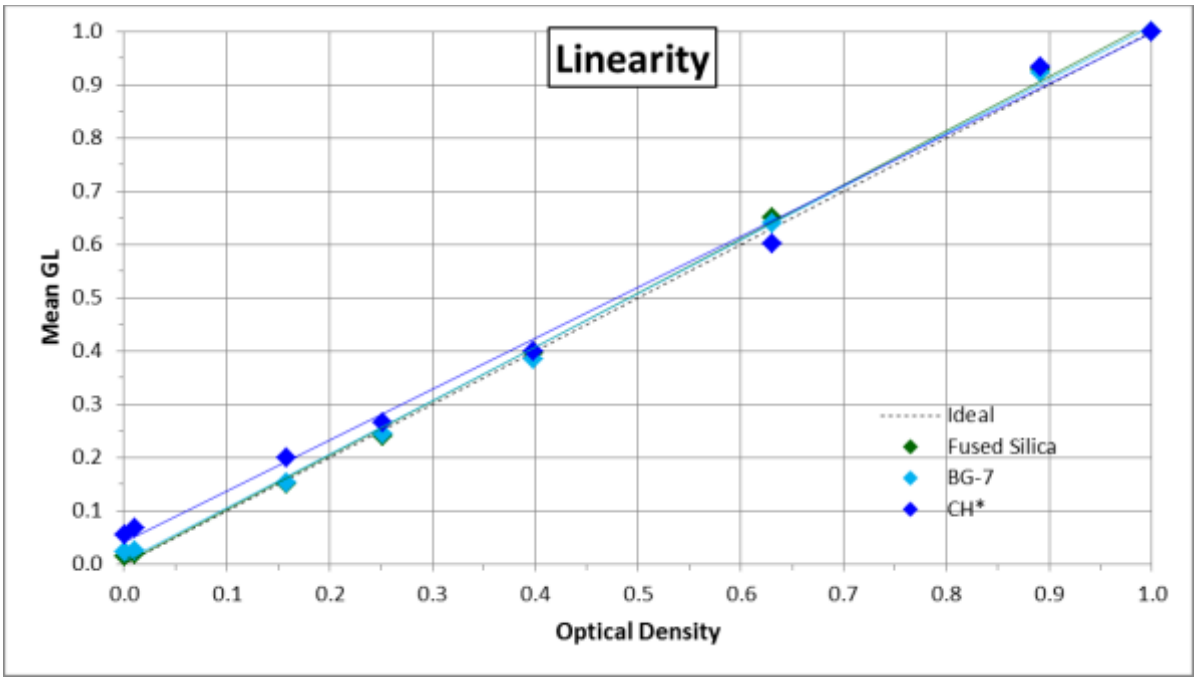


FIGURE 12. Linearity Curves vs. Ideal

5.3 Flat Field / Uniformity

5.3.1 Setup

The Data Camera Package was aligned in front of the 20" Integrating Sphere (as shown in FIGURE 5) with no target in place. The illumination source used was the QTH Lamp (3000K), with a motorized shutter to vary the output intensity. The illumination was set to a Gray Level without saturating the image for each of the 4 filter locations (CH*, Fused Silica, BG-7 & Formaldehyde), resulting in a different illumination level for each filter.

Specific camera settings for this section are as follows (grayed out indicates not applicable):

AVT Camera Functions			Motor Control Functions		
Width/Height (# pixels)	Width	1360	Filter(s) Used	CH*	10,000
	Height	1024		Fused Silica	120,000
Full Frame or Binning	Horiz	1x		Formaldehyde	235,000
	Vert	1x		BG-7	350,000
White Balance	R / B	0.9 / 3.0	Iris	Open	0
Gain	db	0	FOV (mm)	Zoom	Focus
Exposure Time	Full Frame @ 30 fps	32,000	127 x 95.6	50,000	8,400
Pixel Format (Bit Depth)	12 Bit - Raw	BayerRG12			

All images are in the Raw format of the camera (**.bin** file extension), unless otherwise specified.

5.3.2 Data / Analysis

Once the Illumination source was set, 10 images were taken with the same settings. This was repeated for each of the 4 filters.

Using ImageJ, these images were analyzed as a 10-image "stack" resulting in single image with an average intensity. Uniformity data was taken from 2 line profiles, one spanning the horizontal direction and one spanning the vertical direction, providing Min & Max Gray Levels for each direction. The resulting data is shown below, along with the curves in both the horizontal direction (H) and vertical direction (V) corresponding to each filter.

TABLE V. Uniformity Data

Filter	H/V	Max	Min	% Dev
FS	H	3490	2799	19.8%
(50K Avg)	V	3468	3126	9.9%
BG7	H	1809	1450	19.8%
(50K Avg)	V	1798	1611	10.4%
CH	H	1225	893	27.1%
(50K Avg)	V	1193	1027	13.9%
Form	H	1365	1059	22.4%
(50K Avg)	V	1359	1204	11.4%

The curves below resulting from the above data show similar uniformity profiles in both the vertical and horizontal directions. Drop-off at the edges is typical for imaging based systems and is the purpose for this type of characterization. The amplitudes vary due to the adjustment of the illumination for each filter in order to account for transmission differences.

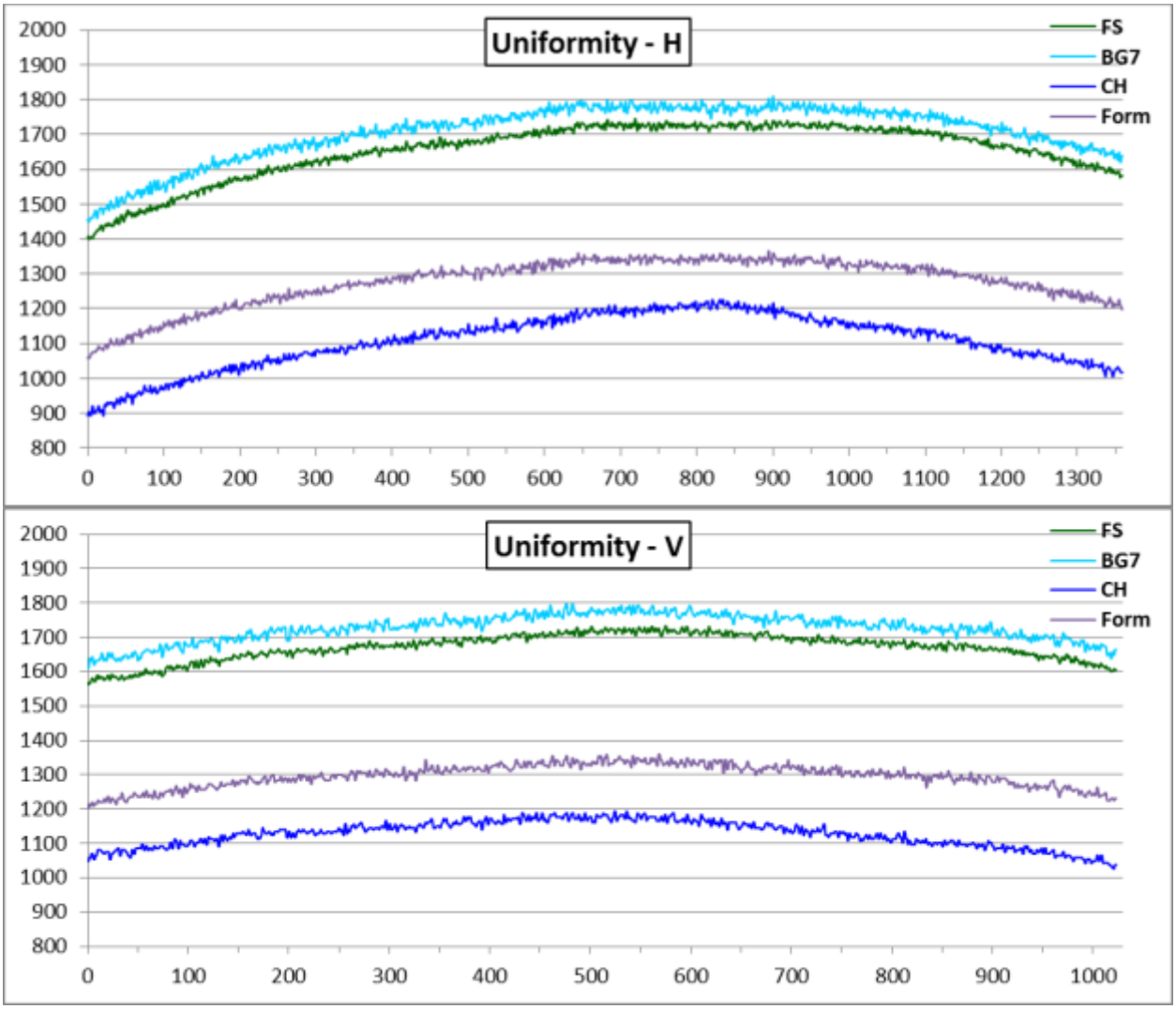


FIGURE 13. Uniformity – Horizontal and Vertical

5.4 Spatial Resolution

5.4.1 Setup

The Data Camera Package was aligned in front of the 20” Integrating Sphere (as shown in FIGURE 5) with a USAF 1951 Target positioned using z-translation, at the specified Object Distance of 253.2 mm from the outer surface of the Data Camera Assembly. The target was centered in the image using x-y adjustments. See image of the target below.

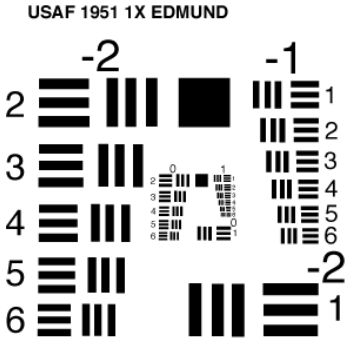


FIGURE 14. USAF 1951 Target

The illumination source used was the QTH Lamp (3000K), with a motorized shutter to vary the output intensity. The illumination was set to maximize the Gray Level without saturating the image for the each of the 4 filter locations. The Spatial Resolution measurement is dependent on filter selection, thus all 4 of the filters (BG7, Formaldehyde, CH, & Fused Silica) were used for this section.

The Spatial Resolution is dependent on the Field of View, thus a wide range of Zoom positions were used for this analysis. See Zoom position correlation to FOV in Section 5.6.

Specific camera settings for this section are as follows (grayed out indicates not applicable):

AVT Camera Functions			Motor Control Functions		
Width/Height (# pixels)	Width	1360	Filter(s) Used	CH*	10,000
	Height	1024		Fused Silica	120,000
Full Frame or Binning	Horiz	1x		Formaldehyde	235,000
	Vert	1x		BG-7	350,000
White Balance	R / B	0.9 / 3.0	Iris	Open	0
Gain	db	0	FOV (mm)	Zoom	Focus
Exposure Time	Full Frame @ 30 fps	32,000	Vary	Vary	Vary
Pixel Format (Bit Depth)	12 Bit - Raw	BayerRG12			

All images are in the Raw format of the camera (**.bin** file extension), unless otherwise specified.

5.4.2 Data / Analysis

The following steps were repeated for each of the 4 filters: Once the illumination was set for each filter, images of the USAF target were captured at Zoom settings spanning the range of Zoom motor counts from 0 to 80,000. See example images below at 0 counts, 40,000 counts, and 80,000 counts. (Note: As the images zoom out to wide field, the black masking on the periphery is due to the size of the illumination output port in the distance and not due to a limitation of the imaging capability.)

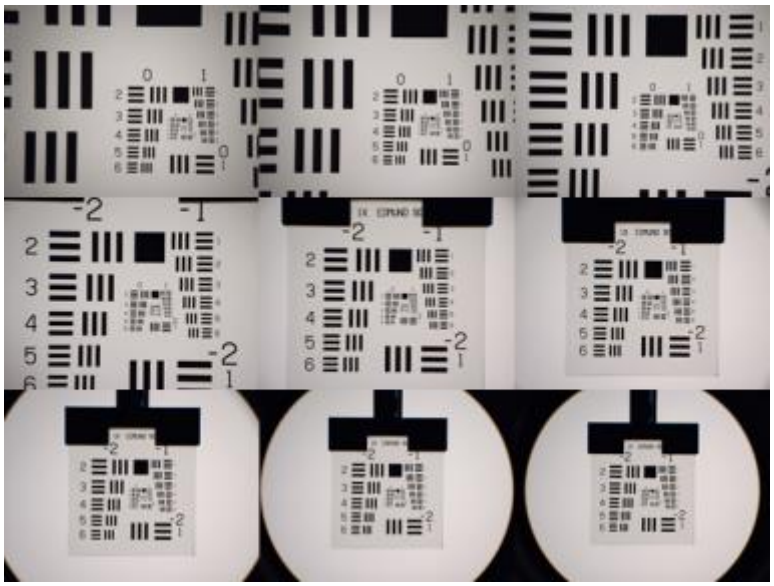


FIGURE 15. USAF Images w/ Fused Silica Filter at 9 Zoom positions

Images were opened using ImageJ imaging software. On the USAF Target, the resolution "limit" is the last group of 3 bars that can be distinguished as such, in both horizontal and vertical directions. Breaks within a bar or the overall pattern are acceptable as long as 3 bars can still be identified. For example, using the "Line Profile" on the image below, the limit where 3 Bars are distinguishable (i.e. still have sufficient contrast) was USAF Group 2, Element 1, corresponding to 3.6 lp/mm (140 μm). This was repeated for each image.

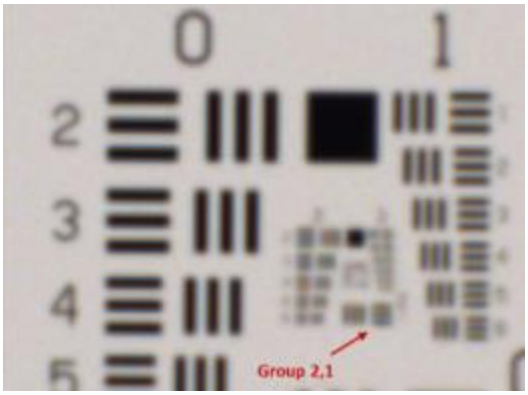


FIGURE 16. Resolution at Group 2,1 on USAF 1951 Target

The following Table lists the Spatial Resolution results for the Data Camera Package over the range of Zoom positions for all 4 of the filters.

TABLE VI. Spatial Resolution Data vs. Zoom position

Fused Silica	Zoom Pos.	Group	lp/mm
Horiz	5000 (35.3 x 26.6)	3,5	12.70
Vert		3,5	12.70
Horiz	10000 (40.9 x 30.8)	3,4	11.31
Vert		3,4	11.31
Horiz	20000 (54.9 x 41.3)	3,2	8.98
Vert		3,2	8.98
Horiz	30000 (74.5 x 56.1)	2,6	7.13
Vert		2,6	7.13
Horiz	40000 (99.1 x 74.6)	2,4	5.66
Vert		2,4	5.66
Horiz	50000 (127.0 x 95.6)	2,1	4.00
Vert		2,1	4.00
Horiz	60000 (152.7 x 114.9)	1,5	3.18
Vert		1,5	3.18
Horiz	70000 (172.5 x 129.8)	1,4	2.83
Vert		1,4	2.83
Horiz	80000 (184.7 x 139.0)	1,3	2.52
Vert		1,3	2.52

BG-7	Zoom Pos.	Group	lp/mm
Horiz	5000 (35.3 x 26.6)	3,4	11.31
Vert		3,4	11.31
Horiz	10000 (40.9 x 30.8)	3,3	10.08
Vert		3,3	10.08
Horiz	20000 (54.9 x 41.3)	3,1	8.00
Vert		3,1	8.00
Horiz	30000 (74.5 x 56.1)	2,5	6.35
Vert		2,5	6.35
Horiz	40000 (99.1 x 74.6)	2,3	5.04
Vert		2,3	5.04
Horiz	50000 (127.0 x 95.6)	1,6	3.56
Vert		1,6	3.56
Horiz	60000 (152.7 x 114.9)	1,5	3.18
Vert		1,5	3.18
Horiz	70000 (172.5 x 129.8)	1,4	2.83
Vert		1,4	2.83
Horiz	80000 (184.7 x 139.0)	1,3	2.52
Vert		1,3	2.52

CH*	Zoom Pos.	Group	lp/mm
Horiz	5,000	3,3	10.08
Vert		3,3	10.08
Horiz	10,000	3,2	8.98
Vert		3,2	8.98
Horiz	20,000	2,6	7.13
Vert		2,6	7.13
Horiz	30,000	2,4	5.66
Vert		2,4	5.66
Horiz	40,000	2,1	4.00
Vert		2,1	4.00
Horiz	50,000	1,4	2.83
Vert		1,4	2.83
Horiz	60,000	1,3	2.52
Vert		1,3	2.52
Horiz	70,000	1,2	2.25
Vert		1,2	2.25
Horiz	80,000	1,1	2.00
Vert		1,1	2.00

Formaldehyde	Zoom Pos.	Group	lp/mm
Horiz	5,000	3,4	11.31
Vert		3,4	11.31
Horiz	10,000	3,3	10.08
Vert		3,3	10.08
Horiz	20,000	3,1	8.00
Vert		3,1	8.00
Horiz	30,000	2,4	5.66
Vert		2,4	5.66
Horiz	40,000	2,1	4.00
Vert		2,1	4.00
Horiz	50,000	1,4	2.83
Vert		1,4	2.83
Horiz	60,000	1,3	2.52
Vert		1,3	2.52
Horiz	70,000	1,2	2.25
Vert		1,2	2.25
Horiz	80,000	1,1	2.00
Vert		1,1	2.00

5.5 Depth of Field (DOF)

5.5.1 Setup

The Data Camera Package was aligned in front of the 20" Integrating Sphere (as shown in FIGURE 5) with a USAF 1951 Target (see FIGURE 14) positioned using z-translation, at the specified Object Distance of 253.2 mm from the outer surface of the Data Camera Assembly. The target was centered in the image using x-y adjustments. This is considered the "Center" position. For Depth of Field measurements, the optical rail was used for depth translation both toward the camera ("–" direction) and away from the camera ("+" direction).

The illumination source used was the QTH Lamp (3000K), with a motorized shutter to vary the output intensity. The illumination was set to maximize the Gray Level without saturating the image for the Fused Silica filter location. The Depth of Field characterization was performed only using the Fused Silica filter location. Two representative Zoom positions (5,000 and 50,000) were used for this analysis.

Specific camera settings for this section are as follows (grayed out indicates not applicable):

AVT Camera Functions			Motor Control Functions		
Width/Height (# pixels)	Width	1360	Filter(s) Used	CH*	10,000
	Height	1024		Fused Silica	120,000
Full Frame or Binning	Horiz	1x		Formaldehyde	235,000
	Vert	1x		BG-7	350,000
White Balance	R / B	0.9 / 3.0	Iris	Open	0
Gain	db	0	FOV (mm)	Zoom	Focus
Exposure Time	Full Frame @ 30 fps	32,000	35.3 x 26.6	5,000	15,400
Pixel Format (Bit Depth)	12 Bit - Raw	BayerRG12	127 x 95.6	50,000	7,900

All images are in the Raw format of the camera (**.bin** file extension), unless otherwise specified.

5.5.2 Data / Analysis

Depth of Field refers to the range of distance, centered at the object plane that an image appears in focus. It is typically defined by Spatial Resolution values at the various stages of depth. If accompanying Resolution values are not given at various points throughout the range, as in this case, it is assumed that the Spatial Resolution measured at the object plane will hold throughout the Depth of Field.

Images of the USAF target were captured at the Center position (253.2 mm distance). The USAF target was then translated along the optical rail away from the camera over the range of +5 mm to +25 mm with images taken every 5 mm. Then the USAF target was translated past the center position, toward the camera over the range of -5 mm to -25 mm again taking images every 5 mm. Note: At each position an image was taken without re-focusing, and again after re-focusing.

See the example below for the +5 position where the image on the left remains at the "Center focus" with the resolution falling off to ~ 8.98 lp/mm (Group 3,2) and the image on the right was re-focused maintaining its original 12.70 lp/mm (Group 3,5).



FIGURE 17. DOF Image at +10: Focus Constant vs. Re-Focus

After examining the differences between not re-focusing and subsequently adjusting the focus at each position, it was clear that under the baseline condition with the Iris wide open, for movement in and out of the image plane, re-focus is necessary to maintain the Spatial Resolution established in Section 5.4.

Subsequently images collected from -25 mm all the way out to +25 mm, were re-focused at each position and after analyzing the images in the same manner as for Spatial Resolution, the results shown in TABLE VI were maintained for both Zoom positions tested (5,000 and 50,000).

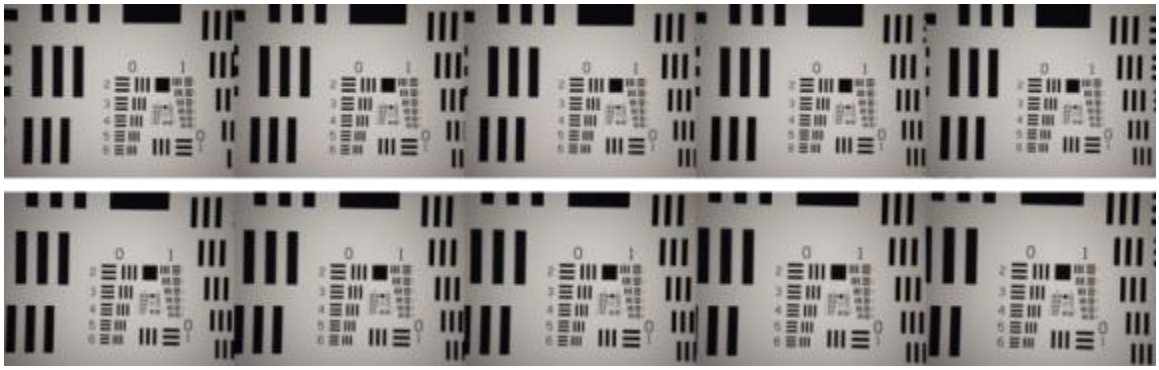


FIGURE 18. Re-focused DOF: +5mm to +25mm (top), -5mm to -25mm (bottom)

Therefore, as the target was translated throughout the -25 mm to +25 mm Depth of Field range, with re-focusing capability, the resolution established at the "0" position was considered to have held at those distances.

5.6 Field of View (FOV)

5.6.1 Setup

The Data Camera Package was aligned in front of the 20" Integrating Sphere (as shown in FIGURE 5) with a Concentric Squares Target positioned using z-translation, at the specified Object Distance of 253.2 mm from the outer surface of the Data Camera Assembly. The target was centered in the image using x-y adjustments. See image of the target below.



FIGURE 19. Concentric Squares Target

The illumination source used was the QTH Lamp (3000K), with a motorized shutter to vary the output intensity. The illumination was set to maximize the Gray Level without saturating the image for the Fused Silica filter location. The Field of View measurements are independent of filter selection, so in this case only the Fused Silica filter was used.

Specific camera settings for this section are as follows (grayed out indicates not applicable):

AVT Camera Functions			Motor Control Functions		
Width/Height (# pixels)	Width	1360	Filter(s) Used	CH*	10,000
	Height	1024		Fused Silica	120,000
Full Frame or Binning	Horiz	1x		Formaldehyde	235,000
	Vert	1x		BG-7	350,000
White Balance	R / B	0.9 / 3.0	Iris	Open	0
Gain	db	0	FOV (mm)	Zoom	Focus
Exposure Time	Full Frame @ 30 fps	32,000	Vary	Vary	Vary
Pixel Format (Bit Depth)	12 Bit - Raw	BayerRG12			

All images are in the Raw format of the camera (**.bin** file extension), unless otherwise specified.

5.6.2 Data / Analysis

Using the setup described above, images of the Concentric Squares target were captured at various Zoom settings, spanning the range of Zoom motor counts from 0 to 80,000. See example images below at 0 counts, 40,000 counts, and 80,000 counts.



FIGURE 20. FOV Images (at Min, Mid, Max Counts)

The images were opened using ImageJ imaging software. Using the "Scale" tool, the line length (in pixels) was stretched spanning the largest full square visible on the target. The pixel value of that line was scaled to the actual square dimension. Then using that scaled value, the "Line Tool" was stretched to span the entire image in both the vertical (1024 pixels) and horizontal (1360 pixels) directions to determine the overall Field of View (FOV) at that Zoom setting. This was repeated for each image, establishing FOV over the Zoom range. See the results table below:

TABLE VII. FOV vs. Zoom Data

Horiz (mm)	Vert (mm)	Zoom (Counts)	Focus (Counts)
31.6	23.8	0	15500
35.3	26.6	5000	15400
40.9	30.8	10000	15250
47.6	35.9	15000	15050
54.9	41.3	20000	14750
64.0	48.2	25000	14325
74.5	56.1	30000	13600
87.1	65.6	35000	12650
99.1	74.6	40000	11500
112.7	84.8	45000	10000
127.0	95.6	50000	8400
141.6	106.6	55000	6900
152.7	114.9	60000	5700
164.5	123.9	65000	4700
172.5	129.8	70000	4150
179.8	135.3	75000	3900
184.7	139.0	80000	3800
188.2	141.8	84000	3700

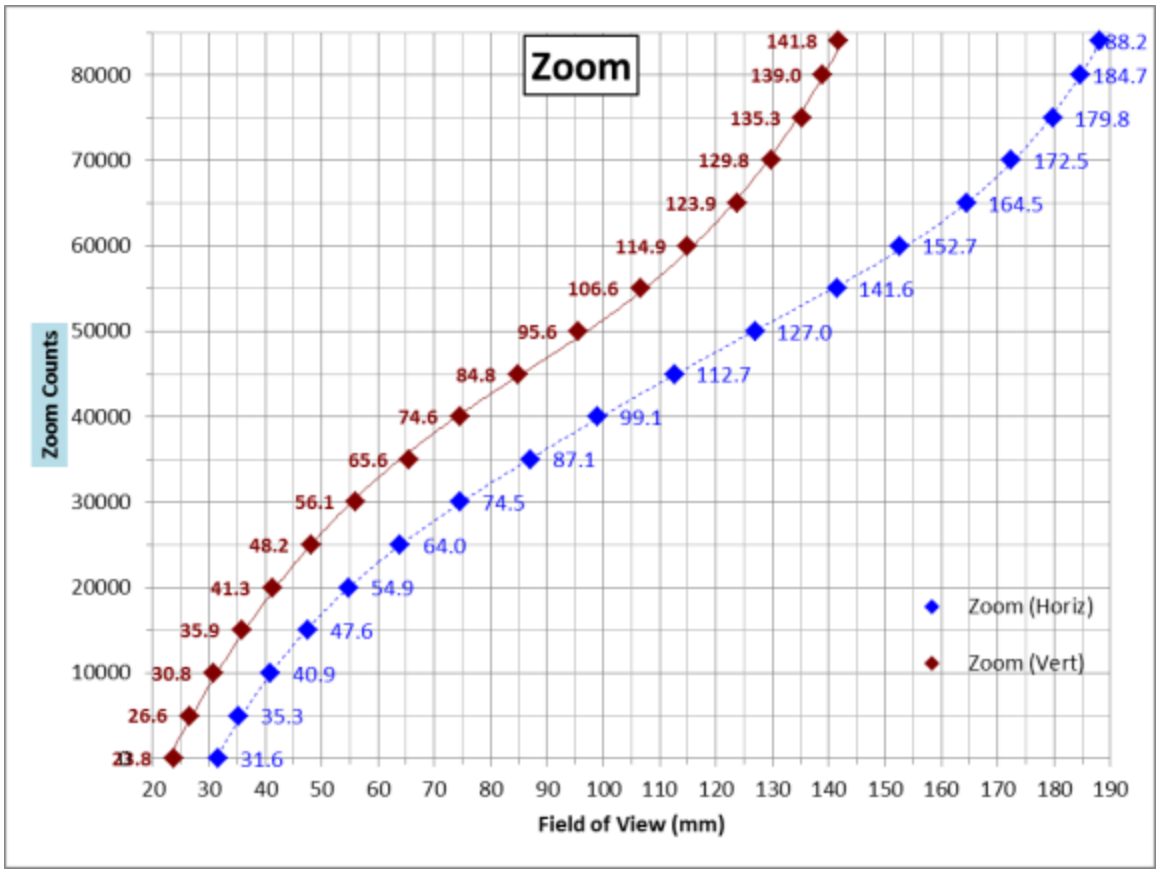


FIGURE 21. FOV vs. Zoom Curve

5.7 Sensitivity – Exposure Time

5.7.1 Setup

The Data Camera Package was aligned in front of the 20" Integrating Sphere (as shown in FIGURE 5) with no target in place. The illumination source used was the QTH Lamp (3000K), with a motorized shutter to vary the output intensity. The illumination was set to maximize the Gray Level without saturating the image for the Fused Silica filter location. This location has the highest transmission and thus would be the first to saturate. Subsequent filters (BG-7 & CH*) were used with the same the illumination setting for a direct comparison. Note that Formaldehyde filter was not characterized due to expected similarity to BG-7 results.

Specific camera settings for this section are as follows (grayed out indicates not applicable):

AVT Camera Functions			Motor Control Functions		
Width/Height (# pixels)	Width	1360	Filter(s) Used	CH*	10,000
	Height	1024		Fused Silica	120,000
Full Frame or Binning	Horiz	1x		Formaldehyde	235,000
	Vert	1x		BG-7	350,000
White Balance	R / B	0.9 / 3.0	Iris	Open	0
Gain	db	0	FOV (mm)	Zoom	Focus
Exposure Time	Full Frame @ 30 fps	Vary	35.3 x 23.8	5,000	15,400
Pixel Format (Bit Depth)	12 Bit - Raw	BayerRG12	Smaller FOV for more Uniform Background		

All images are in the Raw format of the camera (**.bin** file extension), unless otherwise specified.

5.7.2 Data / Analysis

Once the Illumination source was set, the exposure times were varied for each filter, starting with 32,000 µs (32 ms), which is the base setting for 30 frames per second (exp must be < 1/fps).

Images were recorded at each exposure time. These images were analyzed, using ImageJ, over the full 12-bit image (GL from 0 – 4095), providing the Mean Gray Level (GL) for each exposure time. The resulting data is shown below, along with the curves corresponding to each filter. Note the curves are linear, as expected, with slopes varying based on relative intensity.

Exposure Time (µs)	F.S. Mean GL	BG-7 Mean GL	CH* Mean GL
32000	3034.6	1490.0	810.3
30000	2851.6	1397.7	760.7
28000	2668.3	1304.9	711.6
26000	2484.6	1212.8	662.2
24000	2299.6	1120.5	613.0
22000	2111.2	1027.8	563.6
20000	1925.1	935.9	515.2
18000	1735.3	844.1	467.1
16000	1548.3	752.4	418.8
14000	1358.0	661.1	370.9
12000	1166.6	565.4	323.6
10000	975.2	479.6	276.6
8000	784.0	391.0	229.7
6000	594.6	302.3	184.2
4000	406.6	215.4	139.3
2000	222.9	129.8	95.4
1000	133.2	89.6	74.3
500	91.5	70.3	65.3
100	59.6	56.7	56.8

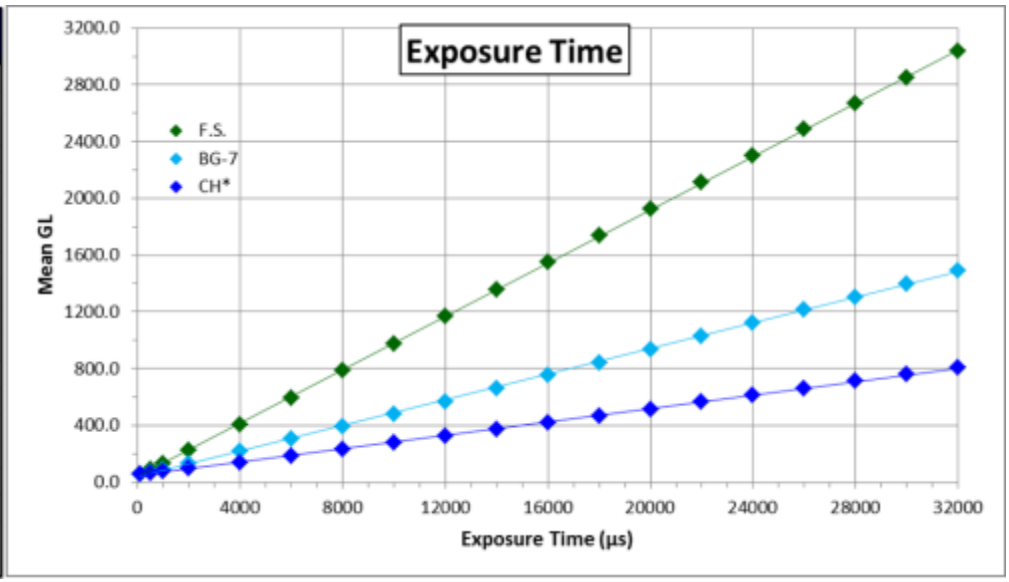


FIGURE 22. Exposure Time (Camera) vs. Mean Gray Levels (GL)

5.8 Sensitivity – Iris

5.8.1 Setup

The Data Camera Package was aligned in front of the 20" Integrating Sphere (as shown in FIGURE 5) with no target in place. The illumination source used was the QTH Lamp (3000K), with a motorized shutter to vary the output intensity. The illumination was set to maximize the Gray Level without saturating the image for the Fused Silica filter location. This location has the highest transmission and thus would be the first to saturate. The Iris measurements are independent of filter selection, so in this case only the Fused Silica filter was characterized.

Specific camera settings for this section are as follows (grayed out indicates not applicable):

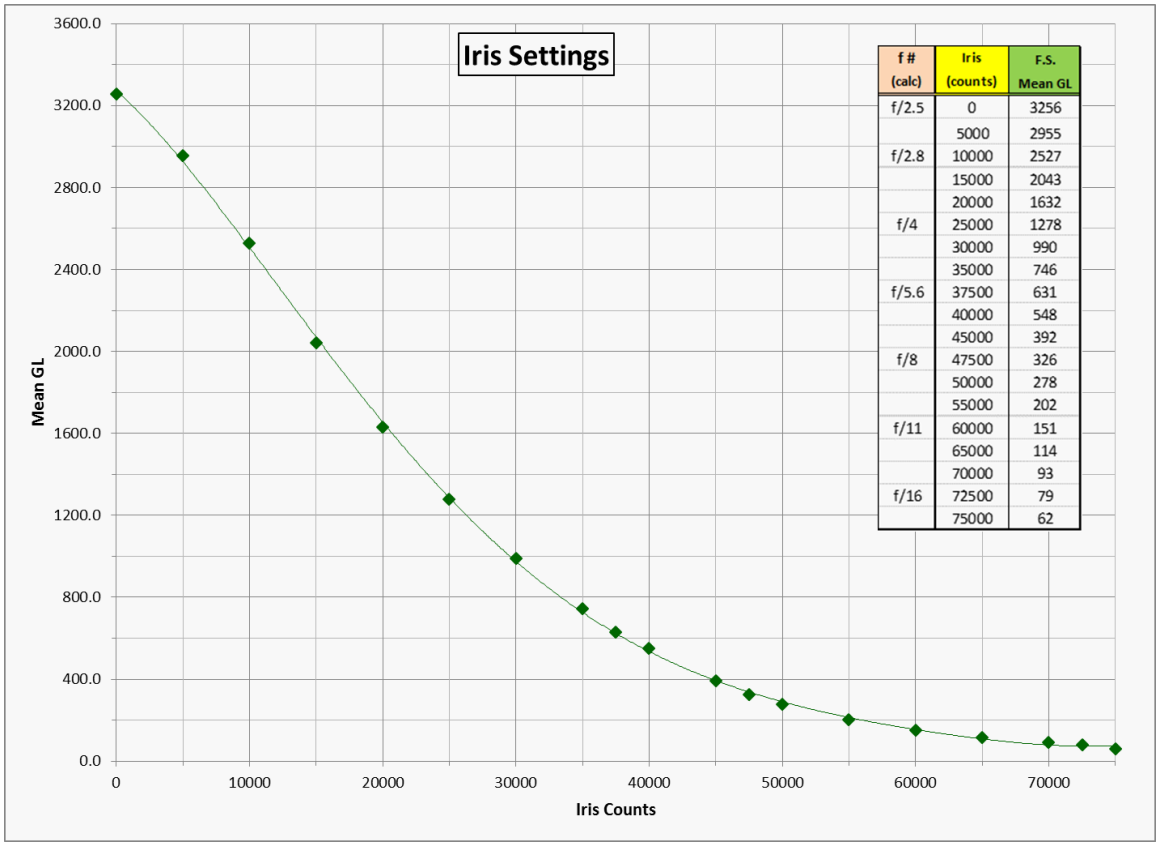
AVT Camera Functions			Motor Control Functions		
Width/Height (# pixels)	Width	1360	Filter(s) Used	CH*	10,000
	Height	1024		Fused Silica	120,000
Full Frame or Binning	Horiz	1x		Formaldehyde	235,000
	Vert	1x		BG-7	350,000
White Balance	R / B	0.9 / 3.0	Iris	Vary	Vary
Gain	db	0	FOV (mm)	Zoom	Focus
Exposure Time	Full Frame @ 30 fps	32,000	35.3 x 23.8	5,000	15,400
Pixel Format (Bit Depth)	12 Bit - Raw	BayerRG12	Smaller FOV for more Uniform Background		

All images are in the Raw format of the camera (**.bin** file extension), unless otherwise specified.

5.8.2 Data / Analysis

Once the Illumination source was set, the Iris counts were varied, starting with 0 (wide open), up to 75,000 (upper limit).

Images were recorded at each Iris setting. These images were analyzed, using ImageJ, over the full 12-bit image (GL from 0 – 4095), providing the Mean Gray Level (GL) for each Iris setting. The resulting data is shown below, along with the corresponding curve. The f-number range in the lens spec is f/2.8 – f/22. Aperture f-number values were calculated based on ½ the Mean GL of the previous f-number.



f # (calc)	Iris (counts)	F.S. Mean GL
f/2.5	0	3256
	5000	2955
f/2.8	10000	2527
	15000	2043
	20000	1632
f/4	25000	1278
	30000	990
	35000	746
f/5.6	37500	631
	40000	548
	45000	392
f/8	47500	326
	50000	278
	55000	202
f/11	60000	151
	65000	114
	70000	93
f/16	72500	79
	75000	62

FIGURE 23. Iris Counts (Lens) vs. Mean Gray Levels (GL)

5.9 Spectral Information

5.9.1 Data

The Spectral information of the ACME Data Camera Package starts with the CCD itself. The Spectral curves for the Sony ICX-285 CCD are shown below. Note the IR cutoff filter is in place, limiting the camera's response above 700 nm.

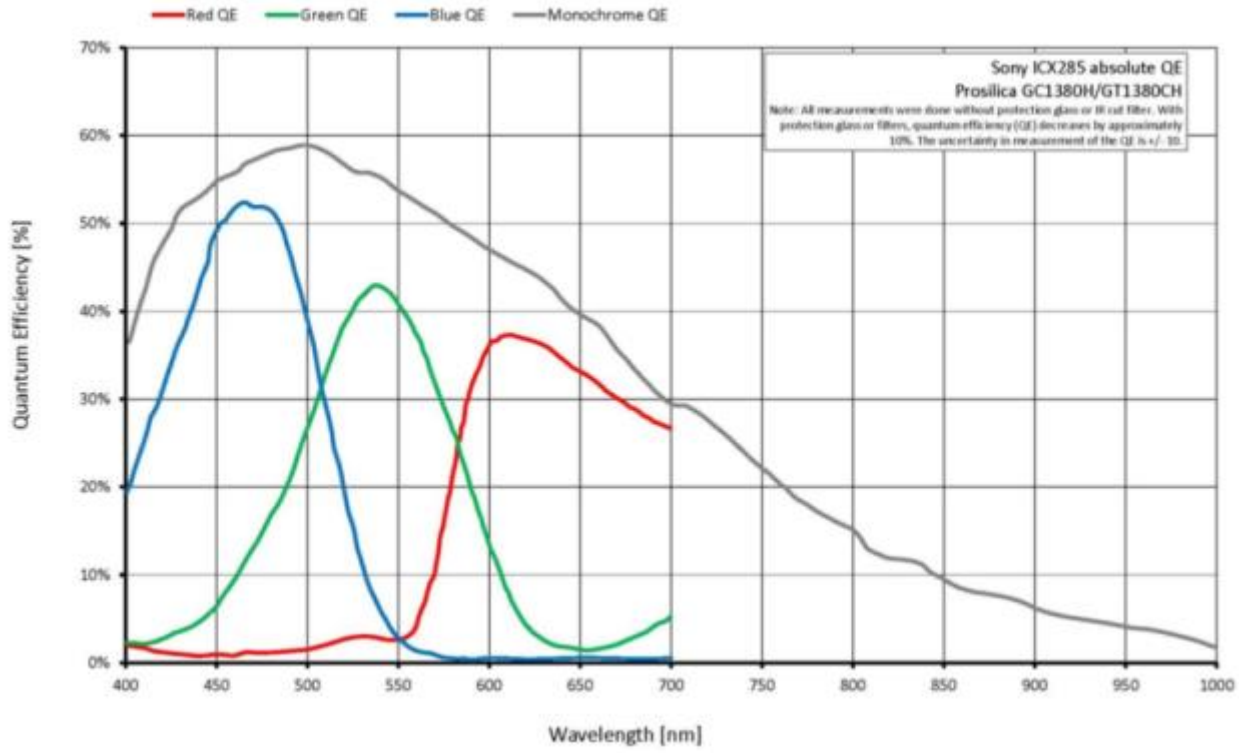


FIGURE 24. Sony ICX-285 Spectral Response

In addition, the Data Camera Package has 4 selectable Filters via the Filter Barrel Assembly: CH* emissions (431 nm ± 10 nm), UV Fused Silica, Formaldehyde emissions (390 nm – 490 nm) and Schott BG-7. The Spectral curves for these Filters are shown below:

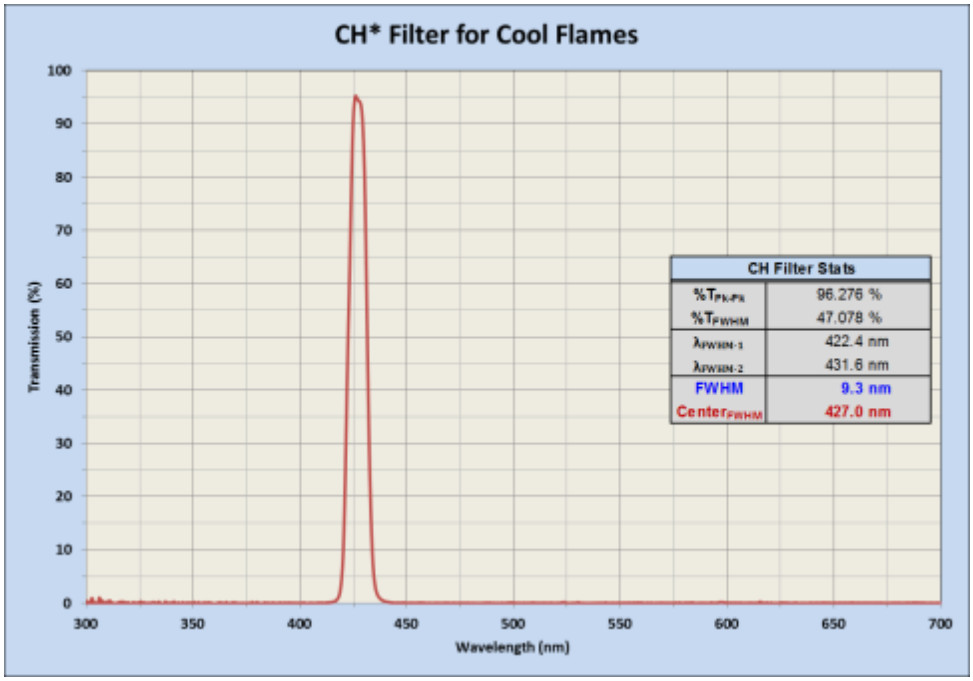


FIGURE 25. CH Filter Transmission

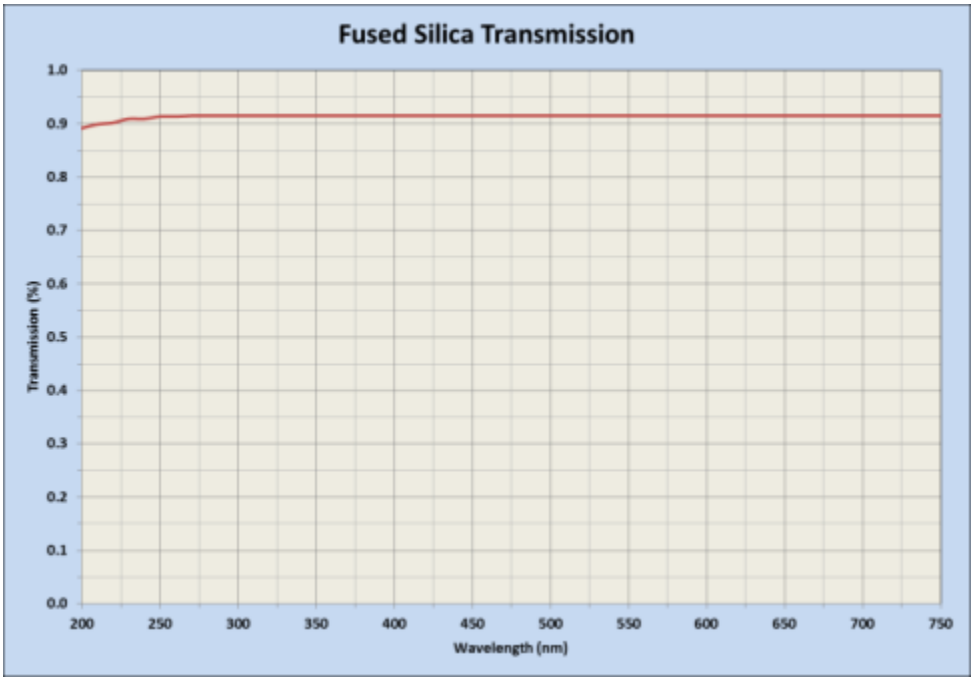


FIGURE 26. UV Fused Silica Transmission

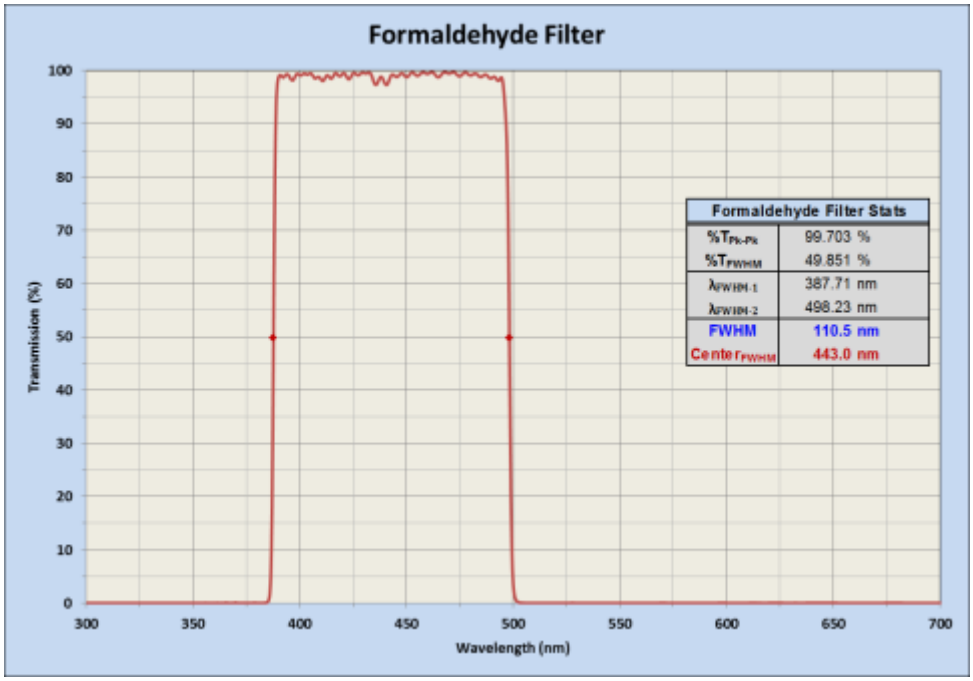


FIGURE 27. Formaldehyde Filter Transmission

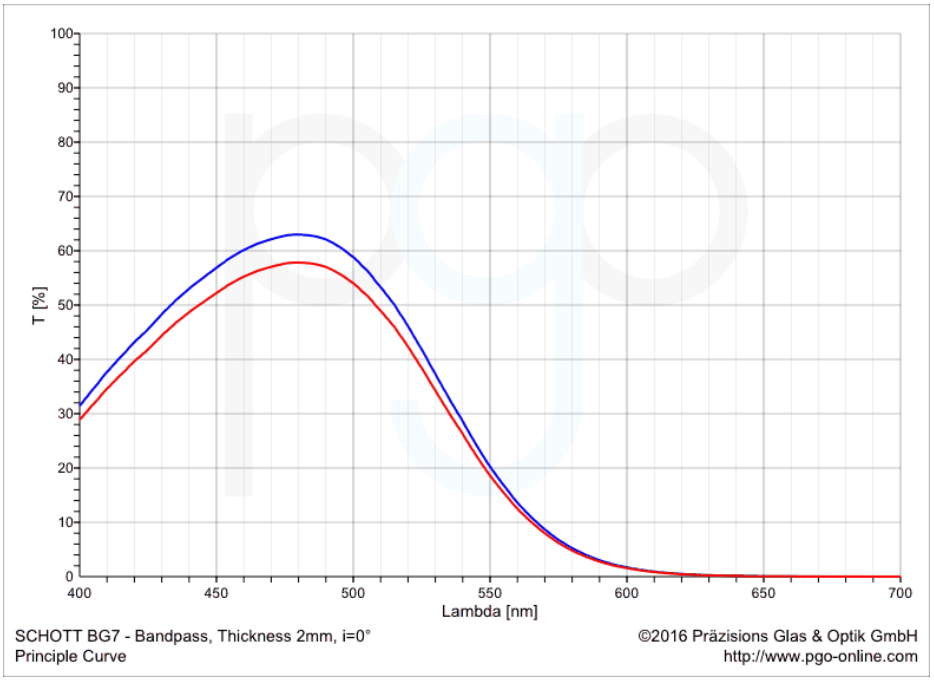


FIGURE 28. BG-7 Filter (2mm) Transmission

The Turning Mirror inside the Filter Barrel Assembly is coated with a UV Enhanced Aluminum Coating. The spectral scan is shown below.

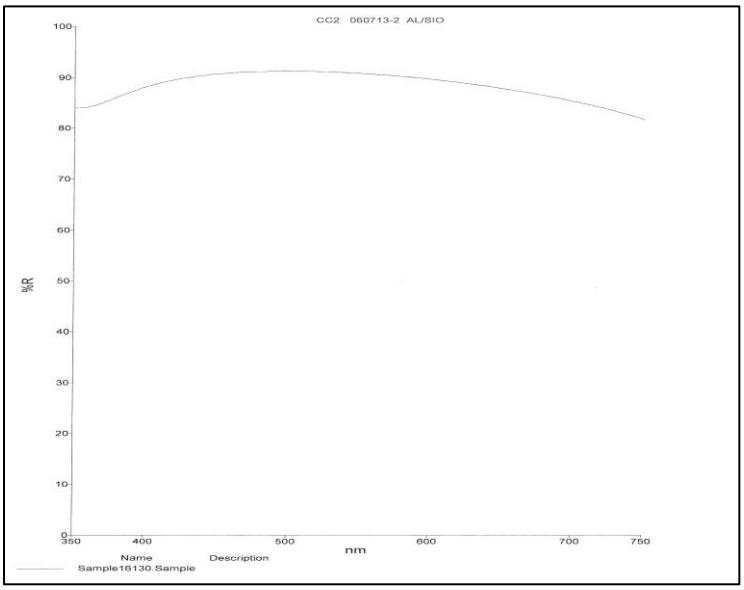


FIGURE 29. UV Enhanced Aluminum Mirror Coating

The final Spectral component in the Data Camera Assembly is the CIR provided 8mm thick Sapphire Window. The transmission data for that is shown below.

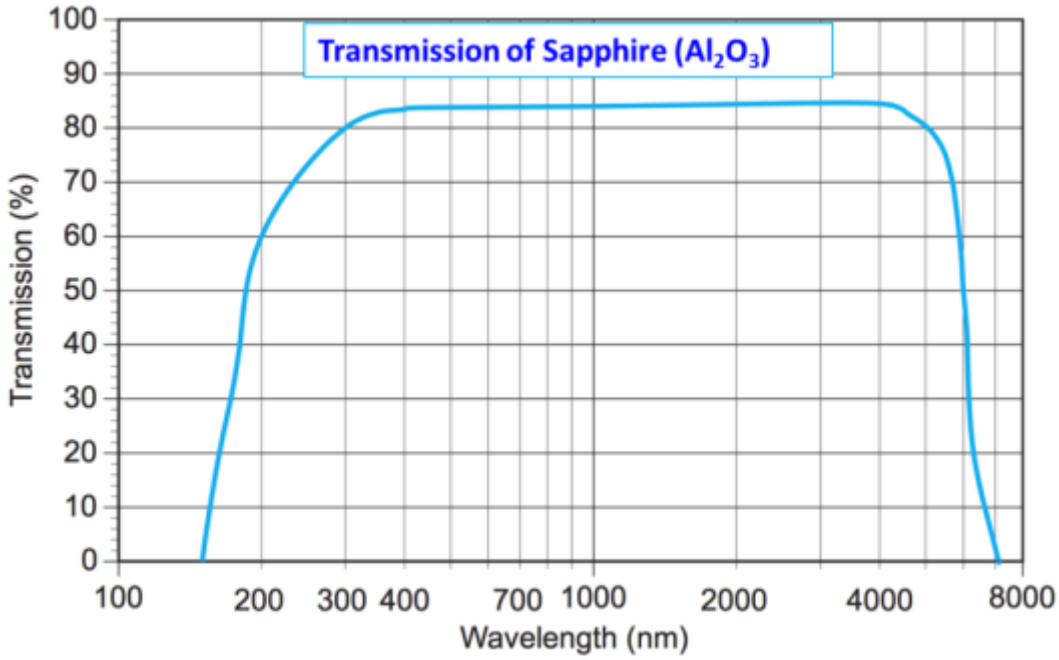


FIGURE 30. CIR Sapphire Window Transmission

5.10 Temporal Stability

5.10.1 Setup

The Data Camera Package was aligned in front of the 20" Integrating Sphere (as shown in FIGURE 5) with no target in place. The illumination source used was the QTH Lamp (3000K), with a motorized shutter to vary the output intensity. The illumination was set to maximize the Gray Level without saturating the image for the CH* filter location. This filter location was chosen since it has the narrowest bandpass, thus would be more sensitive to temporal shifts, so in this case only the CH* filter was characterized.

Specific camera settings for this section are as follows (grayed out indicates not applicable):

AVT Camera Functions			Motor Control Functions		
Width/Height (# pixels)	Width	1360	Filter(s) Used	CH*	10,000
	Height	1024		Fused Silica	120,000
Full Frame or Binning	Horiz	1x		Formaldehyde	235,000
	Vert	1x		BG-7	350,000
White Balance	R / B	0.9 / 3.0	Iris	Open	0
Gain	db	0	FOV (mm)	Zoom	Focus
Exposure Time	Full Frame @ 30 fps	32,000	35.3 x 23.8	5,000	15,400
Pixel Format (Bit Depth)	12 Bit - Raw	BayerRG12	Smaller FOV for more Uniform Background		

All images are in the Raw format of the camera (**.bin** file extension), unless otherwise specified.

5.10.2 Data / Analysis

Once the Illumination source was set and allowed to warm up for 30 minutes, using the CH* filter, images were recorded every minute up to 5 minutes, then in 5 minute intervals through 60 minutes. These images were analyzed, using ImageJ, over the full 12-bit image (GL from 0 – 4095), providing the Mean Gray Level (GL) for each image over time. The resulting data is shown below.

CH* Filter 12-Bit Image	Time (min)	Min	Max	Mean	StdDev	% Diff
CH_Temp_0	0	48	3453	796.7	1128.4	0.3%
CH_Temp_1	1	40	3489	797.1	1128.5	0.4%
CH_Temp_2	2	46	3447	796.9	1128.5	0.3%
CH_Temp_3	3	45	3432	796.1	1127.4	0.2%
CH_Temp_4	4	43	3447	795.7	1126.9	0.2%
CH_Temp_5	5	38	3474	795.1	1126.6	0.1%
CH_Temp_10	10	48	3432	794.6	1125.8	0.1%
CH_Temp_15	15	41	3477	794.4	1125.5	0.0%
CH_Temp_20	20	46	3444	793.6	1124.5	0.1%
CH_Temp_25	25	47	3447	792.8	1123.3	0.2%
CH_Temp_30	30	40	3441	792.8	1122.9	0.2%
CH_Temp_35	35	47	3420	791.7	1121.3	0.3%
CH_Temp_40	40	49	3468	793.2	1123.7	0.1%
CH_Temp_45	45	47	3492	792.6	1122.3	0.2%
CH_Temp_50	50	46	3438	792.8	1122.4	0.2%
CH_Temp_55	55	41	3453	792.0	1121.2	0.3%
CH_Temp_60	60	48	3432	792.5	1122.3	0.2%
Average		44.7	3452	794.15	1124.8	0.2%

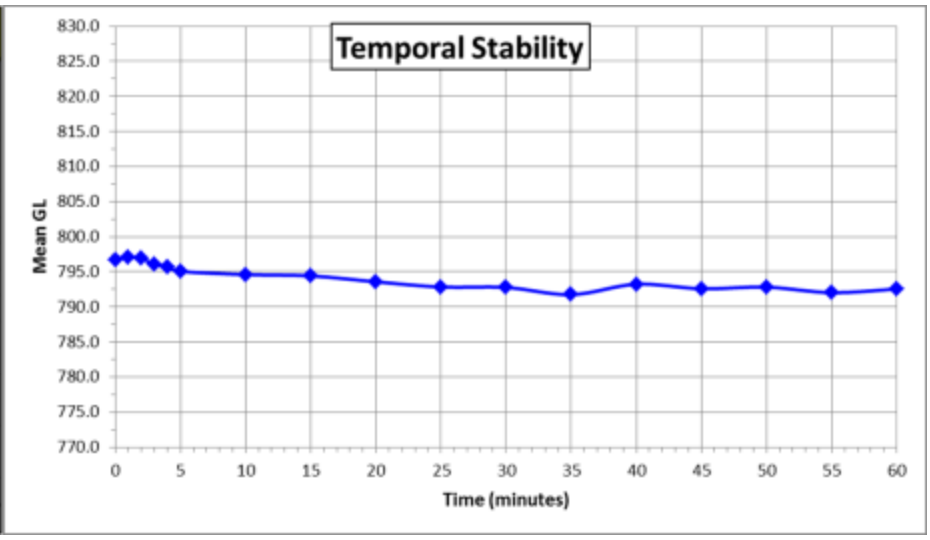


FIGURE 31. Temporal Stability (Mean GL vs. Time)

The average of the Mean GL values was calculated. The deviation of the Mean GL of each image from the overall Mean response was < 0.5%.

5.11 Black Body Characterization (Method 1)

5.11.1 Setup

The Data Camera Package was aligned in front of the Blackbody source (similar to FIGURE 5, but with the Blackbody mounted in place of the integrating sphere). No target is used for this section. The Blackbody source output aperture was set to a diameter of 0.6" (15.24mm) and positioned using z-translation, at the specified Object Distance of 253.2 mm from the outer surface of the Data Camera Assembly. The aperture was centered in the image using x-y adjustments. See image below of the Blackbody source with the Aperture wheel set to 0.6".

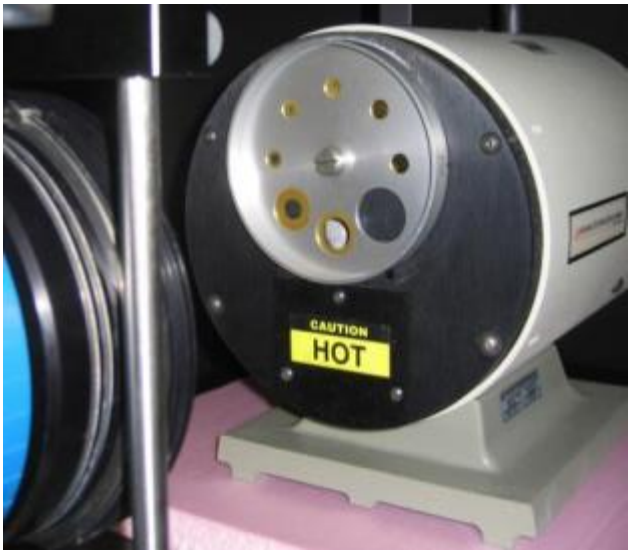


FIGURE 32. Blackbody Source

The Blackbody images are dependent on filter selection, thus all 4 of the filters (BG7, Formaldehyde, CH, & Fused Silica) were used for this section.

Specific camera settings for this section are as follows (grayed out indicates not applicable):

AVT Camera Functions			Motor Control Functions		
Width/Height (# pixels)	Width	1360	Filter(s) Used	CH*	10,000
	Height	1024		Fused Silica	120,000
Full Frame or Binning	Horiz	1x		Formaldehyde	235,000
	Vert	1x		BG-7	350,000
White Balance	R / B	0.9 / 3.0	Iris	Vary	(see table below)
Gain	db	0	FOV (mm)	Zoom	Focus
Exposure Time	Full Frame @ 30 fps	Vary (see table below)	35.3 x 23.8	5,000	15,400
Pixel Format (Bit Depth)	12 Bit - Raw	BayerRG12			

All images are in the Raw format of the camera (**.bin** file extension), unless otherwise specified.

5.11.2 Data / Analysis

In order to prevent the images from saturating, the Blackbody source was set to its highest temperature of 1200°C and for each of the 4 filters, both the Exposure Time and Iris counts were varied until Gray Levels were just under saturation. See the settings below.

TABLE VIII. Exposure Time & Iris Settings for Blackbody Images

Filter	Exposure Time (µs)	Iris
CH*	17,000	17,000
Fused Silica	500	70,000
Formaldehyde	4,100	42,000
BG-7	4,500	28,000

Once the Exposure Time and Iris were set for each filter, images were taken of the Blackbody source aperture at 9 different temperature settings (800°C, 850°C, 900°C, 950°C, 1000°C, 1050°C, 1100°C, 1150°C & 1200°C).

See images below. A subset of temperatures (1000°C to 1200°C) is shown for each filter.

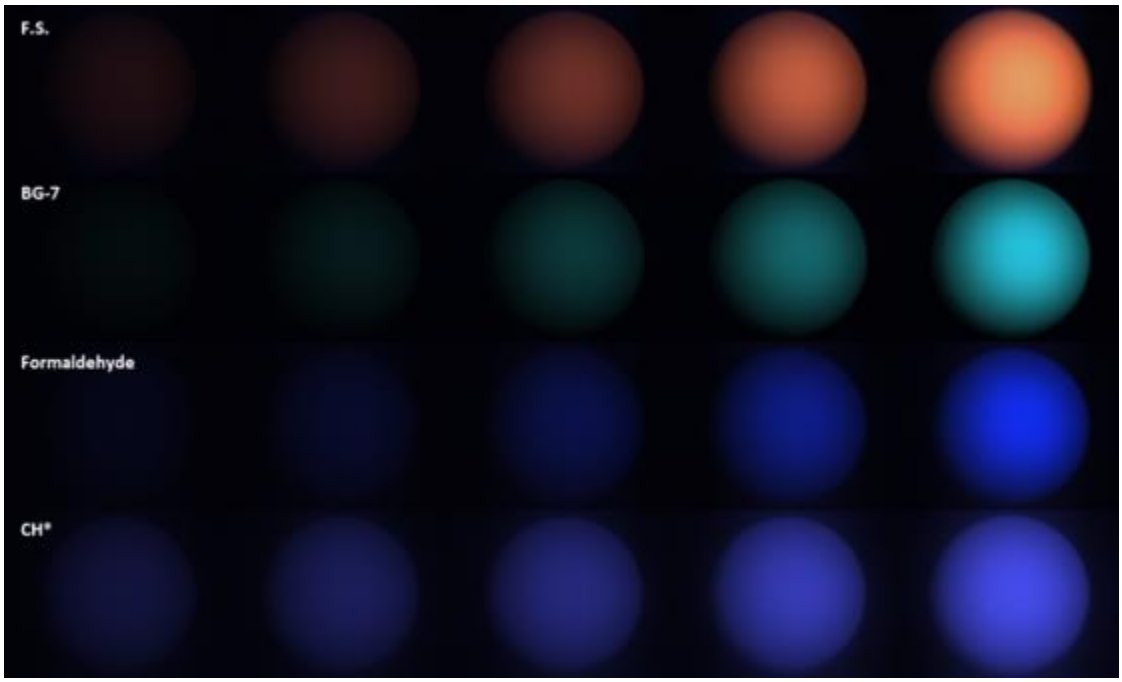


FIGURE 33. Blackbody Images (1000°C – 1200°C) for 4 Filters

These images were analyzed, using ImageJ, over the full 12-bit image (GL from 0 – 4095), providing the Mean Gray Level (GL) for each temperature. The resulting data is shown below, along with the corresponding curves for all 4 filters.

TABLE IX. Blackbody Temp. vs. Mean GL

BB Temp (°C)	F.S. Mean GL	BG-7 Mean GL	CH* Mean GL	Formaldehyde Mean GL
800	32	32	64	32
850	48	32	96	32
900	80	32	144	48
950	144	64	224	80
1000	273	128	337	112
1050	513	273	513	192
1100	931	578	706	305
1150	1702	1075	1043	513
1200	2553	2087	1365	835

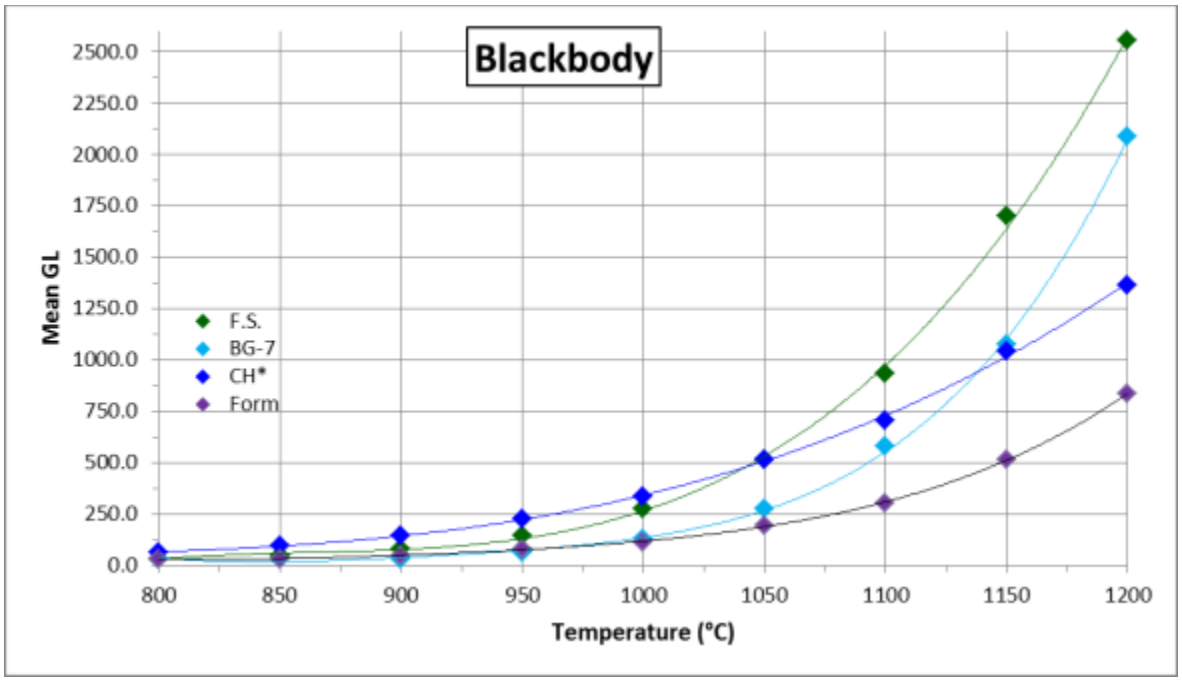


FIGURE 34. Blackbody Temperature vs. Mean GL (4 Filters)

5.12 Platinum Wire & Fiber Images

5.12.1 Setup

The purpose of this section was to obtain images of both the ACME 15 μm fibers and platinum wire, heated to a series of known temperatures (1000°K, 1200°K, 1400°K & 1600°K).

A platinum thermocouple was placed directly next to the wire/fiber being characterized in order to provide direct temperature correlation.

The heating source was a butane burner with a regulator to vary the output. It was mounted on a vertical translation post to provide another method of varying the output at the fiber/wire location. See the setup below.

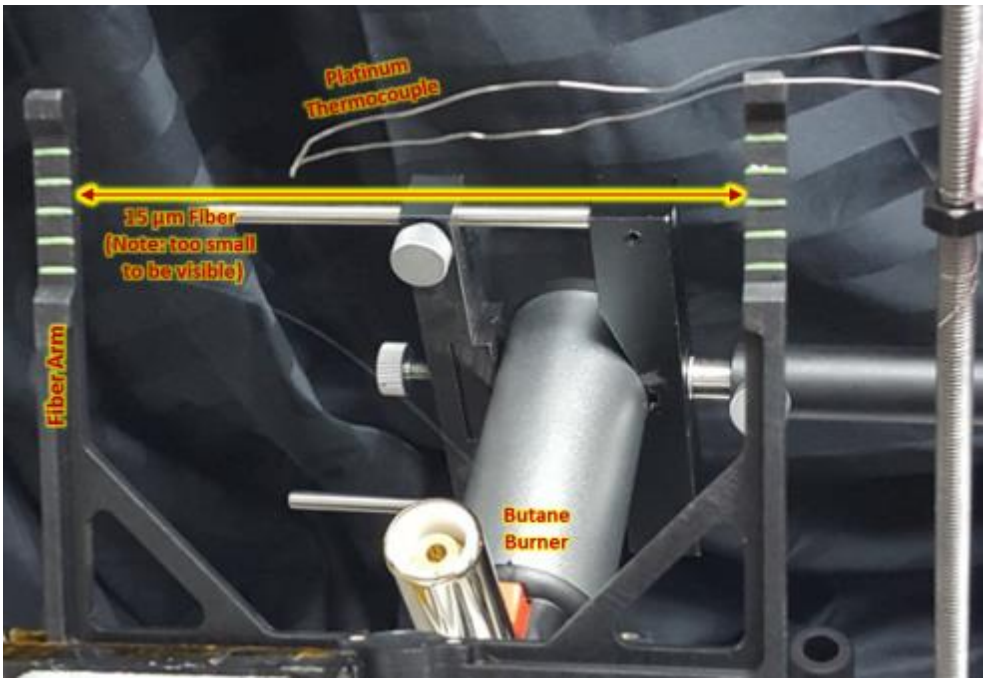


FIGURE 35. Fiber & Platinum Wire Characterization Setup

The Data Camera Package was aligned in front of the fiber and thermocouple, positioned using z-translation, at the specified Object Distance of 253.2 mm from the outer surface of the Data Camera Assembly. See image below of the fiber and thermocouple alignment.

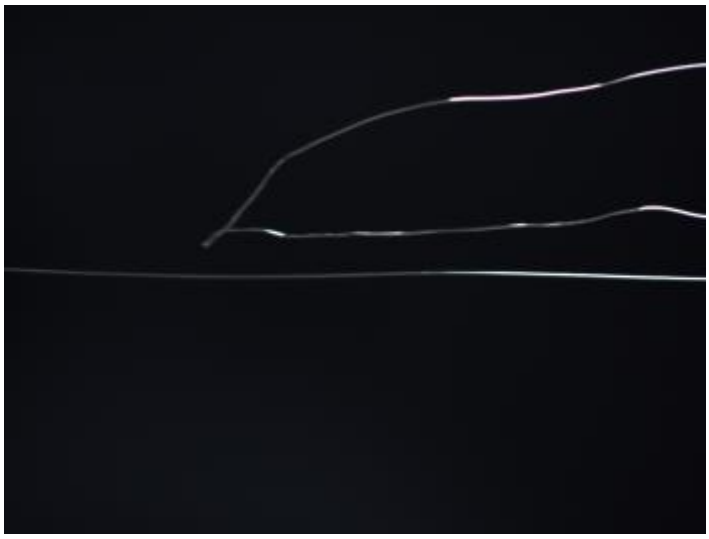


FIGURE 36. Fiber & Platinum Wire Setup Image

The filters used for this characterization were BG7 and Fused Silica).

Specific camera settings for this section are as follows (grayed out indicates not applicable):

AVT Camera Functions			Motor Control Functions		
Width/Height (# pixels)	Width	1360	Filter(s) Used	CH*	10,000
	Height	1024		Fused Silica	120,000
Full Frame or Binning	Horiz	1x		Formaldehyde	235,000
	Vert	1x		BG-7	350,000
White Balance	R / B	0.9 / 3.0	Iris	Open	0
Gain	db	0	FOV (mm)	Zoom	Focus
Exposure Time	Full Frame @ 30 fps	1000 / BG7 500 / FS	35.3 x 23.8	5,000	15,400
Pixel Format (Bit Depth)	12 Bit - Raw	BayerRG12			

All images are in the Raw format of the camera (**.bin** file extension), unless otherwise specified.

5.12.2 Data

Images were saved and provided to the science team for further analysis. An example of one of the images showing the glowing fiber and thermocouple (above) is shown below.



FIGURE 37. Fiber & Platinum Wire Characterization Image

6.0 SECONDARY CHARACTERIZATION RESULTS

6.1 Blackbody Characterization (Method 2)

6.1.1 Setup

The purpose of this section was to obtain images for absolute light calibration. This requires images of a source with a known spectral content and intensity. In this case a Blackbody source was used over a known temperature range of 800°C – 1200°C.

The Data Camera Package was aligned in front of the Blackbody source. No target is used for this section. The Blackbody source output aperture was set to its largest diameter of 1.0" (25.4 mm) and positioned using z-translation, at the specified Object Distance of 253.2 mm from the outer surface of the Data Camera Assembly. The Blackbody aperture was centered in the image using x-y adjustments. See a picture of the Blackbody setup below.



FIGURE 38. Blackbody Characterization Setup

The Blackbody images for this method were taken over a range of temperatures (800°C – 1200°C) with the Exposure Time setting of the camera optimized to use the majority of the dynamic range of the CCD. In order to avoid saturation, the Exposure Time was set for each image condition such that the resulting image had Gray Level counts between 3000 – 3500, equivalent to 80% – 85% of the 12 bit range (0 – 4095).

Once at a particular temperature, images were taken over a range of Field of View settings via changing the Zoom & Focus parameters on the lens.

Also at each FOV, multiple Aperture settings were used (f/2.5, f/5.6 & f/11) via adjustment of the lens Iris. Exposure Time setting was adjusted accordingly with the change in aperture.

This method of Blackbody Characterization was performed on the 2 main filters (BG7 & Fused Silica) intended for use on ACME.

Background images (~10) in the same dark room with the Blackbody at room temperature were taken for each set of [exposure time, zoom, iris & filter] conditions as well.

Specific camera settings for this section are as follows:

AVT Camera Functions		
Width/Height (# pixels)	Width	1360
	Height	1024
Full Frame/Binning (Independent for H/V)	Horiz	1x
	Vert	1x
White Balance (Independent for R/B)	R	1.0
	B	1.0
Exposure Time	Base: 32,000	Vary
Pixel Format	12 Bit - Raw	Bayer16
Filter Positions		
Barrel Position	Name	Counts
Position 2	Fused Silica	120,000
Position 4	BG-7	350,000

All images are in the Raw format of the camera (**.bin or .raw** file extension), unless otherwise specified.

6.1.2 Data / Analysis

In order to use as much of the dynamic range as possible and prevent the images from saturating, the Exposure Time was optimized for each condition. See the Exposure Time (in µs) results for all of the image conditions in the table below.

TABLE X. Exposure Time Values for Blackbody Characterization

Temp. (°C)	Filter	Zoom [cts]	Focus [cts]	Iris	Iris [cts]	Exposure [µs]
800	FS	0	17200	2.5	0	90,000
				5.6	37500	450,000
				11	60000	2,050,000
		30000	13450	2.5	0	85,000
				5.6	37500	450,000
				11	60000	2,250,000
		50000	8250	2.5	0	85,000
				5.6	37500	450,000
				11	60000	2,500,000
		80000	3750	2.5	0	85,000
				5.6	37500	460,000
				11	60000	2,500,000
850	FS	0	17200	2.5	0	36,000
				5.6	37500	185,000
				11	60000	950,000
		30000	13450	2.5	0	35,000
				5.6	37500	185,000
				11	60000	950,000
		50000	8250	2.5	0	35,000
				5.6	37500	185,000
				11	60000	950,000
		80000	3750	2.5	0	35,000
				5.6	37500	185,000
				11	60000	955,000
900	FS	0	17200	2.5	0	16,000
				5.6	37500	78,000
				11	60000	425,000
		30000	13450	2.5	0	14,500
				5.6	37500	78,000
				11	60000	425,000
		50000	8250	2.5	0	14,500
				5.6	37500	78,000
				11	60000	425,000
		80000	3750	2.5	0	14,500
				5.6	37500	78,000
				11	60000	425,000
950	FS	0	17200	2.5	0	7,150
				5.6	37500	36,000
				11	60000	195,000
		30000	13450	2.5	0	6,900
				5.6	37500	36,000
				11	60000	195,000
		50000	8250	2.5	0	6,900
				5.6	37500	36,000
				11	60000	195,000
		80000	3750	2.5	0	6,900
				5.6	37500	36,000
				11	60000	195,000
1000	FS	0	17200	2.5	0	3,400
				5.6	37500	17,500
				11	60000	90,000
		30000	13450	2.5	0	3,250
				5.6	37500	17,500
				11	60000	90,000
		50000	8250	2.5	0	3,250
				5.6	37500	17,500
				11	60000	91,000
		80000	3750	2.5	0	3,250
				5.6	37500	17,500
				11	60000	91,000

Temp. (°C)	Filter	Zoom [cts]	Focus [cts]	Iris	Iris [cts]	Exposure [µs]
800	BG7	0	17200	2.5	0	4,000,000
				5.6	37500	8,000,000
				11	60000	8,000,000
		30000	13450	2.5	0	3,750,000
				5.6	37500	8,000,000
				11	60000	8,000,000
		50000	8250	2.5	0	3,750,000
				5.6	37500	8,000,000
				11	60000	8,000,000
		80000	3750	2.5	0	3,750,000
				5.6	37500	8,000,000
				11	60000	8,000,000
850	BG7	0	17200	2.5	0	1,350,000
				5.6	37500	5,800,000
				11	60000	800,000
		30000	13450	2.5	0	1,250,000
				5.6	37500	7,250,000
				11	60000	8,000,000
		50000	8250	2.5	0	1,250,000
				5.6	37500	7,250,000
				11	60000	8,000,000
		80000	3750	2.5	0	1,250,000
				5.6	37500	7,250,000
				11	60000	8,000,000
900	BG7	0	17200	2.5	0	500,000
				5.6	37500	2,600,000
				11	60000	8,000,000
		30000	13450	2.5	0	475,000
				5.6	37500	2,600,000
				11	60000	8,000,000
		50000	8250	2.5	0	475,000
				5.6	37500	2,600,000
				11	60000	8,000,000
		80000	3750	2.5	0	475,000
				5.6	37500	2,600,000
				11	60000	8,000,000
950	BG7	0	17200	2.5	0	200,000
				5.6	37500	1,050,000
				11	60000	4,750,000
		30000	13450	2.5	0	185,000
				5.6	37500	1,050,000
				11	60000	5,750,000
		50000	8250	2.5	0	185,000
				5.6	37500	1,050,000
				11	60000	5,750,000
		80000	3750	2.5	0	185,000
				5.6	37500	1,050,000
				11	60000	5,750,000
1000	BG7	0	17200	2.5	0	87,500
				5.6	37500	445,000
				11	60000	2,400,000
		30000	13450	2.5	0	82,500
				5.6	37500	445,000
				11	60000	2,400,000
		50000	8250	2.5	0	82,500
				5.6	37500	445,000
				11	60000	2,400,000
		80000	3750	2.5	0	82,500
				5.6	37500	445,000
				11	60000	2,400,000

Temp. (°C)	Filter	Zoom [cts]	Focus [cts]	Iris	Iris [cts]	Exposure [µs]
1050	FS	0	17200	2.5	0	1,750
				5.6	37500	9,000
				11	60000	47,500
		30000	13450	2.5	0	1,650
				5.6	37500	9,000
				11	60000	47,500
		50000	8250	2.5	0	1,650
				5.6	37500	9,000
				11	60000	47,500
		80000	3750	2.5	0	1,650
				5.6	37500	9,000
				11	60000	47,500
1100	FS	0	17200	2.5	0	900
				5.6	37500	4,750
				11	60000	25,000
		30000	13450	2.5	0	850
				5.6	37500	4,750
				11	60000	25,000
		50000	8250	2.5	0	850
				5.6	37500	4,750
				11	60000	25,000
		80000	3750	2.5	0	850
				5.6	37500	4,750
				11	60000	25,000
1150	FS	0	17200	2.5	0	500
				5.6	37500	2,600
				11	60000	13,750
		30000	13450	2.5	0	475
				5.6	37500	2,600
				11	60000	13,750
		50000	8250	2.5	0	475
				5.6	37500	2,600
				11	60000	13,750
		80000	3750	2.5	0	475
				5.6	37500	2,600
				11	60000	13,750
1200	FS	0	17200	2.5	0	275
				5.6	37500	1,500
				11	60000	8,000
		10000	15900	2.5	0	260
				5.6	37500	1,500
				11	60000	8,000
		20000	15250	2.5	0	260
				5.6	37500	1,500
				11	60000	8,000
		30000	13450	2.5	0	270
				5.6	37500	1,500
				11	60000	8,000
		40000	11500	2.5	0	270
				5.6	37500	1,500
				11	60000	8,000
		50000	8250	2.5	0	270
				5.6	37500	1,500
				11	60000	8,000
		60000	5500	2.5	0	270
				5.6	37500	1,500
				11	60000	8,000
		70000	4000	2.5	0	270
				5.6	37500	1,500
				11	60000	8,000
		80000	3750	2.5	0	270
				5.6	37500	1,500
				11	60000	8,000

Temp. (°C)	Filter	Zoom [cts]	Focus [cts]	Iris	Iris [cts]	Exposure [µs]
1050	BG7	0	17200	2.5	0	39,000
				5.6	37500	200,000
				11	60000	1,100,000
		30000	13450	2.5	0	36,000
				5.6	37500	200,000
				11	60000	1,100,000
		50000	8250	2.5	0	36,000
				5.6	37500	200,000
				11	60000	1,100,000
		80000	3750	2.5	0	36,000
				5.6	37500	200,000
				11	60000	1,100,000
1100	BG7	0	17200	2.5	0	18,000
				5.6	37500	102,000
				11	60000	425,000
		30000	13450	2.5	0	17,000
				5.6	37500	101,000
				11	60000	525,000
		50000	8250	2.5	0	17,000
				5.6	37500	101,000
				11	60000	525,000
		80000	3750	2.5	0	17,000
				5.6	37500	101,000
				11	60000	525,000
1150	BG7	0	17200	2.5	0	9,500
				5.6	37500	49,000
				11	60000	260,000
		30000	13450	2.5	0	9,000
				5.6	37500	49,000
				11	60000	260,000
		50000	8250	2.5	0	8,900
				5.6	37500	49,000
				11	60000	260,000
		80000	3750	2.5	0	8,900
				5.6	37500	49,000
				11	60000	265,000
1200	BG7	0	17200	2.5	0	5,000
				5.6	37500	26,000
				11	60000	140,000
		10000	15900	2.5	0	4,750
				5.6	37500	26,000
				11	60000	140,000
		20000	15250	2.5	0	4,750
				5.6	37500	26,000
				11	60000	140,000
		30000	13450	2.5	0	4,750
				5.6	37500	26,000
				11	60000	140,000
		40000	11500	2.5	0	4,750
				5.6	37500	26,000
				11	60000	140,000
		50000	8250	2.5	0	4,750
				5.6	37500	26,000
				11	60000	140,000
		60000	5500	2.5	0	4,750
				5.6	37500	26,000
				11	60000	140,000
		70000	4000	2.5	0	4,750
				5.6	37500	26,000
				11	60000	140,000
		80000	3750	2.5	0	4,750
				5.6	37500	26,000
				11	60000	140,000

6.2 Thermocouple and Flame Characterization

6.2.1 Setup

The purpose of this section was to obtain images of a steady flame with a platinum thermocouple in that flame, allowing it to be imaged to a series of known temperatures (1250°C, 1400°C, 1600°C, 1650°C, 1700°C, 1750°C, 1800°C and 1900°C).

The heating source was a propane based burner with a regulator and diffusor at the exit port to precisely control the output. The burner was mounted on a vertical translation post in order to provide height adjustment relative to the thermocouple location.

The cylindrical junction of the S-type platinum thermocouple was placed directly at the base of the flame in order to provide direct temperature correlation.

The Data Camera Package was aligned in front of the thermocouple junction. The thermocouple junction was positioned using z-translation, at the specified Object Distance of 253.2 mm from the outer surface of the Data Camera Assembly. See an image of the setup below.

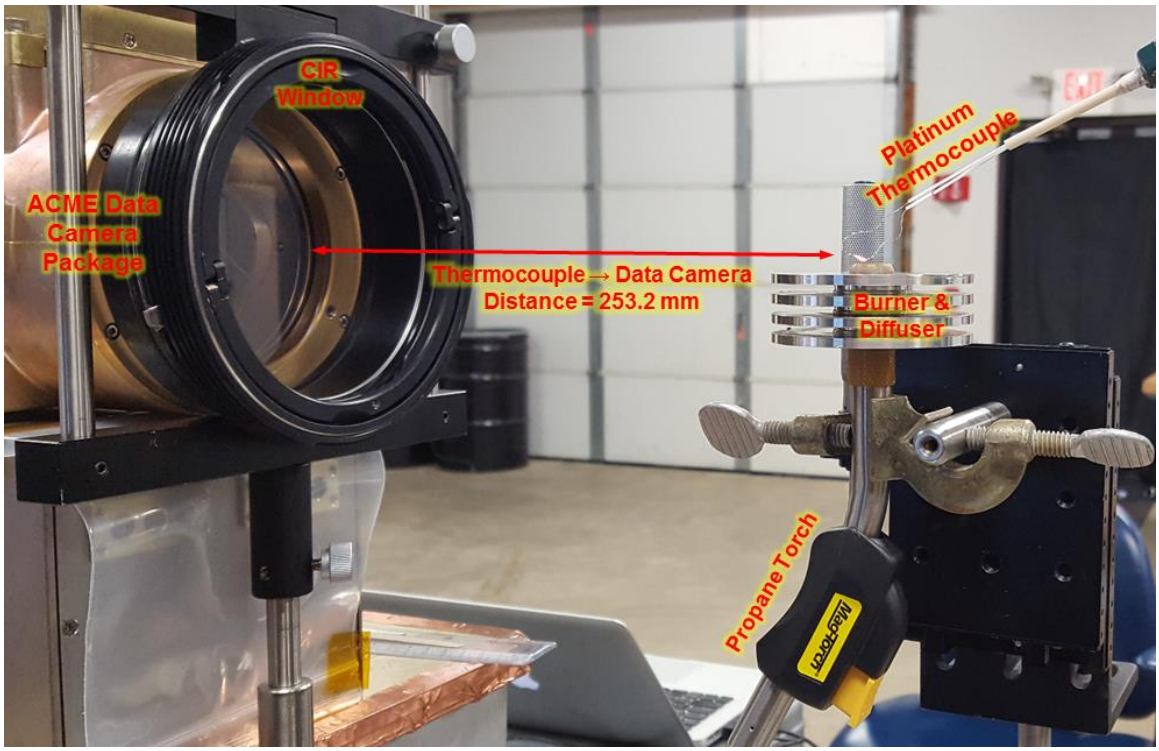


FIGURE 39. Thermocouple Characterization Setup

For more detailed information about the specific components in the propane based system used to produce the flame, along with operating instructions and safety information, see the "ACME Data Camera Characterization Test Plan & Procedure," ACME-TST-061.

The images of the thermocouple were taken over a range of temperatures (1250°C, 1400°C, 1600°C, 1650°C, 1700°C, 1750°C, 1800°C and 1900°C) obtained by means of adjusting flow of fuel to the flame. See a setup image (not from the ACME Data Camera) of the flame and thermocouple below.

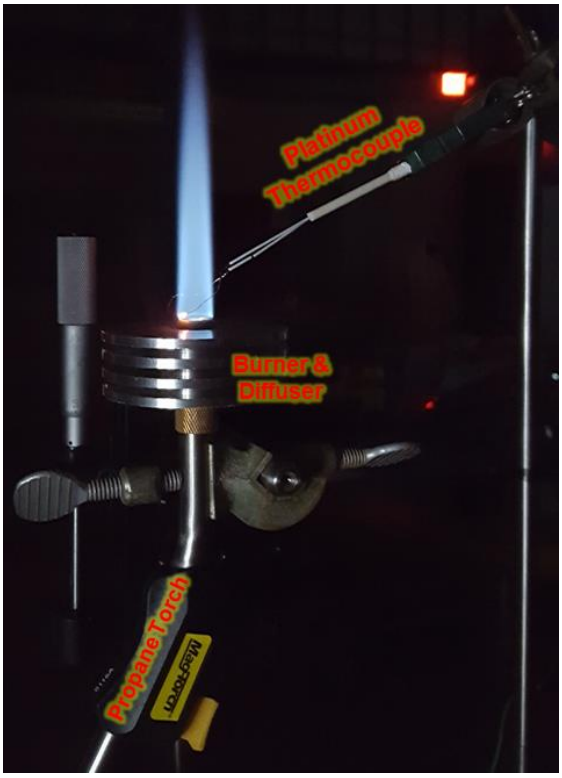


FIGURE 40. Thermocouple with Flame

For each set of images, the Exposure Time setting of the camera was optimized to use the majority of the dynamic range of the CCD. In order to avoid saturation, the Exposure Time was set for each image condition such that the resulting image had Gray Level counts between 3000 – 3500, equivalent to 80% – 85% of the 12 bit range (0 – 4095).

Once at a particular temperature, images were taken over a range of Field of View settings via changing the Zoom & Focus parameters on the lens.

Also at each FOV, multiple Aperture settings were used (f/2.5, f/5.6 & f/11) via adjustment of the lens Iris. Exposure Time setting was adjusted accordingly with the change in aperture.

This thermocouple characterization was performed only on the BG7 filter.

Background images (~10) in the same dark room with the thermocouple and burner at room temperature were taken for each set of [exposure time, zoom & iris] conditions as well.

Specific camera settings for this section are as follows:

AVT Camera Functions		
Width/Height (# pixels)	Width	1360
	Height	1024
Full Frame/Binning (Independent for H/V)	Horiz	1x
	Vert	1x
White Balance (Independent for R/B)	R	1.0
	B	1.0
Exposure Time	Base: 32,000	Vary
Pixel Format	12 Bit - Raw	Bayer16
Filter Position		
Barrel Position	Name	Counts
Position 4	BG-7	350,000

All images are in the Raw format of the camera (**.bin or .raw** file extension), unless otherwise specified.

6.2.2 Data / Analysis

Images were saved and provided to the science team for further analysis. An example of one of the images showing the glowing thermocouple is shown below.

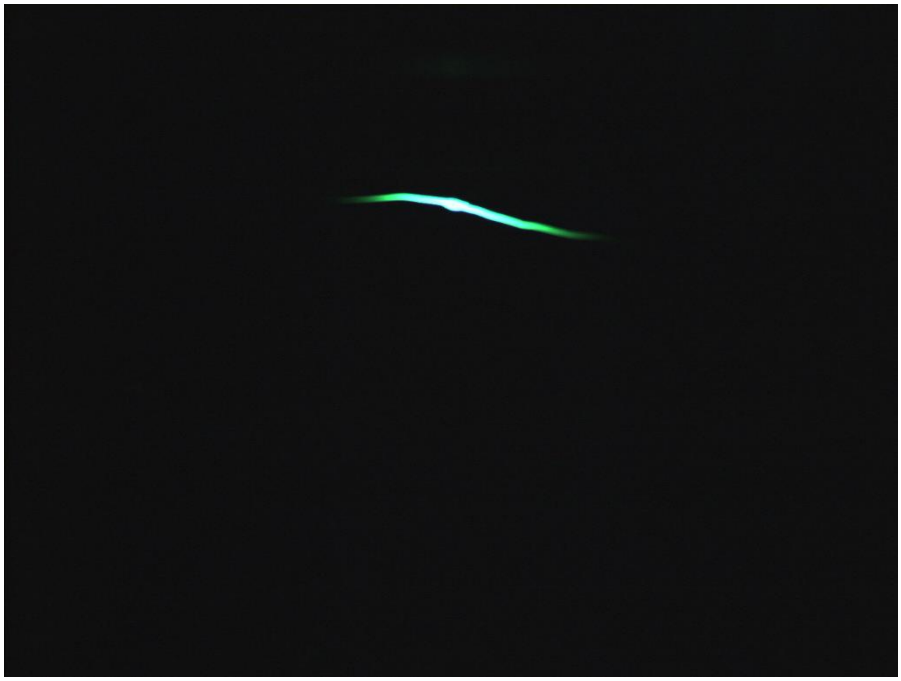


FIGURE 41. Thermocouple Characterization Image

Again, in order to maximize the dynamic range and prevent the images of the thermocouple from saturating, the Exposure Time was optimized for each condition. See the resulting Exposure Time (in µs) for all of the image conditions in the table below.

TABLE XI. Exposure Time Values for Thermocouple Characterization

Temp. (°C)	Filter	Zoom [cts]	Focus [cts]	Iris	Iris [cts]	Exposure [µs]
1250	BG7	0 [max]	16300	2.5	0	250000
				4	25000	500000
				5.6	37500	1000000
				8	47500	2000000
				11	60000	4000000
				16	72500	8000000
		10000	15250	2.5	0	250000
				4	25000	500000
				5.6	37500	1000000
				8	47500	2000000
				11	60000	4000000
				16	72500	8000000
		20000	14750	2.5	0	250000
				4	25000	500000
				5.6	37500	1000000
				8	47500	2000000
				11	60000	4000000
				16	72500	8000000
		30000	13550	2.5	0	150000
				4	25000	500000
				5.6	37500	1000000
				8	47500	2000000
				11	60000	4000000
				16	72500	8000000
1400	BG7	0 [max]	16300	2.5	0	40000
				4	25000	100000
				5.6	37500	200000
				8	47500	400000
				11	60000	800000
				16	72500	2000000
		10000	15250	2.5	0	40000
				4	25000	100000
				5.6	37500	200000
				8	47500	400000
				11	60000	800000
				16	72500	2000000
		20000	14750	2.5	0	40000
				4	25000	100000
				5.6	37500	200000
				8	47500	400000
				11	60000	800000
				16	72500	2000000
		30000	13550	2.5	0	40000
				4	25000	100000
				5.6	37500	200000
				8	47500	400000
				11	60000	800000
				16	72500	2000000

Temp. (°C)	Filter	Zoom [cts]	Focus [cts]	Iris	Iris [cts]	Exposure [µs]
1600	BG7	0 [max]	16300	2.5	0	5000
				4	25000	12000
				5.6	37500	25000
				8	47500	50000
				11	60000	150000
				16	72500	500000
		10000	15250	2.5	0	5000
				4	25000	15000
				5.6	37500	30000
				8	47500	70000
				11	60000	150000
				16	72500	500000
		20000	14750	2.5	0	5000
				4	25000	15000
				5.6	37500	30000
				8	47500	70000
				11	60000	150000
				16	72500	500000
		30000	13550	2.5	0	5000
				4	25000	15000
				5.6	37500	30000
				8	47500	70000
				11	60000	200000
				16	72500	600000
1650	BG7	0 [max]	16300	2.5	0	2500
				4	25000	6000
				5.6	37500	12000
				8	47500	25000
				11	60000	60000
				16	72500	200000
		10000	15250	2.5	0	2200
				4	25000	6000
				5.6	37500	12000
				8	47500	25000
				11	60000	65000
				16	72500	200000
		20000	14750	2.5	0	2200
				4	25000	6000
				5.6	37500	12000
				8	47500	25000
				11	60000	60000
				16	72500	200000
		30000	13550	2.5	0	2500
				4	25000	7000
				5.6	37500	15000
				8	47500	30000
				11	60000	70000
				16	72500	250000

Temp. (°C)	Filter	Zoom [cts]	Focus [cts]	Iris	Iris [cts]	Exposure [µs]
1700	BG7	0 [max]	16300	2.5	0	1500
				4	25000	4000
				5.6	37500	8000
				8	47500	15000
				11	60000	40000
				16	72500	150000
		10000	15250	2.5	0	1600
				4	25000	5000
				5.6	37500	10000
				8	47500	20000
				11	60000	50000
				16	72500	150000
		20000	14750	2.5	0	16000
				4	25000	4500
				5.6	37500	9500
				8	47500	18000
				11	60000	45000
				16	72500	150000
		30000	13550	2.5	0	1600
				4	25000	5000
				5.6	37500	10000
				8	47500	20000
				11	60000	50000
				16	72500	150000
1750	BG7	0 [max]	16300	2.5	0	1000
				4	25000	2500
				5.6	37500	5000
				8	47500	10000
				11	60000	30000
				16	72500	85000
		10000	15250	2.5	0	1000
				4	25000	3000
				5.6	37500	6000
				8	47500	12000
				11	60000	30000
				16	72500	90000
		20000	14750	2.5	0	1000
				4	25000	3000
				5.6	37500	6000
				8	47500	12000
				11	60000	30000
				16	72500	90000
		30000	13550	2.5	0	1000
				4	25000	3000
				5.6	37500	6000
				8	47500	12000
				11	60000	30000
				16	72500	100000

Temp. (°C)	Filter	Zoom [cts]	Focus [cts]	Iris	Iris [cts]	Exposure [µs]
1800	BG7	0 [max]	16300	2.5	0	700
				4	25000	1500
				5.6	37500	3000
				8	47500	7000
				11	60000	15000
				16	72500	60000
		10000	15250	2.5	0	600
				4	25000	1800
				5.6	37500	4000
				8	47500	7000
				11	60000	20000
				16	72500	60000
		20000	14750	2.5	0	600
				4	25000	1800
				5.6	37500	3500
				8	47500	6500
				11	60000	20000
				16	72500	60000
		30000	3700	2.5	0	600
				4	25000	1800
				5.6	37500	3500
				8	47500	7000
				11	60000	20000
				16	72500	60000
1900	BG7	0 [max]	16300	2.5	0	200
				4	25000	500
				5.6	37500	1200
				8	47500	2300
				11	60000	5500
				16	72500	20000
		10000	15250	2.5	0	200
				4	25000	500
				5.6	37500	1200
				8	47500	2300
				11	60000	5500
				16	72500	20000
		20000	14750	2.5	0	180
				4	25000	500
				5.6	37500	1100
				8	47500	2200
				11	60000	5600
				16	72500	20000
		30000	3700	2.5	0	200
				4	25000	600
				5.6	37500	1200
				8	47500	2500
				11	60000	6000
				16	72500	20000

APPENDIX A ACRONYMS AND ABBREVIATIONS

ACME	Advanced Combustion via Microgravity Experiments
BB	Broadband
BG7	Blue-Green 7 Filter (Schott)
CH*	Carbon-Hydrogen Radical
CIR	Combustion Integrated Rack
COTS	Commercial off The Shelf
DC	Direct Current
DOF	Depth of Field
ERD	Engineering Requirements Document
ESD	Electrostatic Discharge
FCF	Fluids and Combustion Facility
Form	Formaldehyde
FOV	Field of View
FS	Fused Silica
GL	Gray Level
IR	Infrared
ISS	International Space Station
N/A	Not Applicable
NASA	National Aeronautics and Space Administration
NB	Narrowband
ND	Neutral Density
RGB	Red, Green, Blue
SNR	Signal to Noise Ratios
SpaceDOC	Space Flight Systems Development and Operations Contract
UV	Ultraviolet
VDC	Volts (direct current)